tropical: SubsetSumDecision

2026-10-25T03:52:34

1 Key results

This report covers 240 runs (5 seeds; 0 diverged) under frozen hash `037f987733ac`.

Where to look. The headline metric is `ood_len_f1` (higher is better). Figure 1 plots it against `fishr_lambda`. Table 1 gives every cell's mean with its 95% bootstrap CI. Table 2 compares each level with the control. Table 3 gives the trend. Table 4 gives the paired contrasts. Its tests take the parametric path: values away from the metric bounds. Cells are compared within a seed: a seed index fixes the data and the initialization. Comparisons are marked significant at $p < 0.05$ after the stated correction.

1.1 `ood_len_f1`

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.3282	[0.252, 0.406]	0.0968	5	0
adaptive	pointer	0.01	0.3224	[0.27, 0.375]	0.0726	5	0
adaptive	pointer	0.1	0.3229	[0.27, 0.376]	0.0724	5	0
adaptive	pointer	1	0.3252	[0.273, 0.373]	0.062	5	0
adaptive	pointer	10	0.3274	[0.274, 0.381]	0.0699	5	0
adaptive	pointer	100	0.3267	[0.277, 0.377]	0.0666	5	0
adaptive	pointer	1000	0.3054	[0.255, 0.365]	0.0724	5	0
adaptive	pointer	1e+04	0.2944	[0.248, 0.347]	0.0653	5	0
adaptive	seq2seq	0	0.05996	[0.0473, 0.0717]	0.0152	5	0
adaptive	seq2seq	0.01	0.06171	[0.0502, 0.0732]	0.0143	5	0
adaptive	seq2seq	0.1	0.0603	[0.0502, 0.0699]	0.0126	5	0
adaptive	seq2seq	1	0.05993	[0.0509, 0.0669]	0.0108	5	0
adaptive	seq2seq	10	0.06558	[0.0522, 0.0789]	0.0177	5	0
adaptive	seq2seq	100	0.06288	[0.0479, 0.0771]	0.0193	5	0
adaptive	seq2seq	1000	0.03155	[0.0207, 0.0418]	0.0141	5	0
adaptive	seq2seq	1e+04	0.01568	[0.00754, 0.0228]	0.0103	5	0
tropical	pointer	0	0.3284	[0.281, 0.376]	0.0589	5	0
tropical	pointer	0.01	0.3535	[0.321, 0.377]	0.0371	5	0
tropical	pointer	0.1	0.3067	[0.293, 0.321]	0.0181	5	0
tropical	pointer	1	0.375	[0.318, 0.423]	0.0682	5	0
tropical	pointer	10	0.3494	[0.301, 0.396]	0.0651	5	0
tropical	pointer	100	0.3868	[0.331, 0.422]	0.0623	5	0
tropical	pointer	1000	0.3556	[0.296, 0.415]	0.0768	5	0
tropical	pointer	1e+04	0.3333	[0.284, 0.383]	0.0639	5	0
tropical	seq2seq	0	0.2945	[0.22, 0.369]	0.0939	5	0
tropical	seq2seq	0.01	0.2174	[0.156, 0.299]	0.0967	5	0
tropical	seq2seq	0.1	0.2543	[0.158, 0.35]	0.121	5	0
tropical	seq2seq	1	0.2479	[0.179, 0.321]	0.092	5	0
tropical	seq2seq	10	0.2608	[0.196, 0.322]	0.0859	5	0
tropical	seq2seq	100	0.1651	[0.135, 0.204]	0.0469	5	0

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	1000	0.1811	[0.114, 0.28]	0.112	5	0
tropical	seq2seq	1e+04	0.02423	[0.00629, 0.0422]	0.0234	5	0
vanilla	pointer	0	0.2391	[0.21, 0.266]	0.0364	5	0
vanilla	pointer	0.01	0.3074	[0.239, 0.372]	0.0863	5	0
vanilla	pointer	0.1	0.302	[0.233, 0.37]	0.087	5	0
vanilla	pointer	1	0.3038	[0.236, 0.375]	0.0896	5	0
vanilla	pointer	10	0.3035	[0.237, 0.371]	0.0844	5	0
vanilla	pointer	100	0.2973	[0.234, 0.363]	0.0836	5	0
vanilla	pointer	1000	0.2996	[0.234, 0.358]	0.0842	5	0
vanilla	pointer	1e+04	0.2933	[0.249, 0.337]	0.0602	5	0
vanilla	seq2seq	0	0.06356	[0.0517, 0.0754]	0.0151	5	0
vanilla	seq2seq	0.01	0.06088	[0.0461, 0.0757]	0.0187	5	0
vanilla	seq2seq	0.1	0.06306	[0.0473, 0.0796]	0.0212	5	0
vanilla	seq2seq	1	0.06108	[0.048, 0.0736]	0.0166	5	0
vanilla	seq2seq	10	0.07919	[0.0607, 0.0945]	0.0229	5	0
vanilla	seq2seq	100	0.07859	[0.0562, 0.0963]	0.0264	5	0
vanilla	seq2seq	1000	0.03947	[0.0338, 0.0452]	0.00741	5	0
vanilla	seq2seq	1e+04	0.01994	[0.0107, 0.0266]	0.0107	5	0

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	-0.005812	[-0.133, 0.122]	1.0000	
adaptive	pointer	0.1	-0.005338	[-0.133, 0.122]	1.0000	
adaptive	pointer	1	-0.003003	[-0.13, 0.124]	1.0000	
adaptive	pointer	10	-8.349×10^{-4}	[-0.128, 0.127]	1.0000	
adaptive	pointer	100	-0.001539	[-0.129, 0.126]	1.0000	
adaptive	pointer	1000	-0.0228	[-0.15, 0.105]	0.9957	
adaptive	pointer	1e+04	-0.03384	[-0.161, 0.0936]	0.9625	
adaptive	seq2seq	0.01	0.00175	[-0.0237, 0.0272]	1.0000	
adaptive	seq2seq	0.1	3.382×10^{-4}	[-0.0251, 0.0258]	1.0000	
adaptive	seq2seq	1	-2.794×10^{-5}	[-0.0255, 0.0254]	1.0000	
adaptive	seq2seq	10	0.00562	[-0.0199, 0.0311]	0.9857	
adaptive	seq2seq	100	0.002922	[-0.0226, 0.0284]	0.9997	
adaptive	seq2seq	1000	-0.02841	[-0.0539, -0.00293]	0.0236	*
adaptive	seq2seq	1e+04	-0.04428	[-0.0698, -0.0188]	0.0002	*
tropical	pointer	0.01	0.02501	[-0.0782, 0.128]	0.9764	
tropical	pointer	0.1	-0.02179	[-0.125, 0.0814]	0.9888	
tropical	pointer	1	0.04655	[-0.0566, 0.15]	0.6969	
tropical	pointer	10	0.02096	[-0.0822, 0.124]	0.9910	
tropical	pointer	100	0.05835	[-0.0449, 0.162]	0.4768	
tropical	pointer	1000	0.02712	[-0.0761, 0.13]	0.9644	
tropical	pointer	1e+04	0.00482	[-0.0984, 0.108]	1.0000	
tropical	seq2seq	0.01	-0.07714	[-0.233, 0.0791]	0.6135	
tropical	seq2seq	0.1	-0.04025	[-0.196, 0.116]	0.9678	
tropical	seq2seq	1	-0.04659	[-0.203, 0.11]	0.9352	
tropical	seq2seq	10	-0.03367	[-0.19, 0.123]	0.9874	
tropical	seq2seq	100	-0.1294	[-0.286, 0.0269]	0.1382	
tropical	seq2seq	1000	-0.1134	[-0.27, 0.0428]	0.2353	
tropical	seq2seq	1e+04	-0.2703	[-0.427, -0.114]	0.0002	*
vanilla	pointer	0.01	0.06833	[-0.0687, 0.205]	0.6035	
vanilla	pointer	0.1	0.06287	[-0.0741, 0.2]	0.6818	
vanilla	pointer	1	0.06468	[-0.0723, 0.202]	0.6558	

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p	
vanilla	pointer	10	0.06443	[-0.0726, 0.201]	0.6593	
vanilla	pointer	100	0.05816	[-0.0788, 0.195]	0.7473	
vanilla	pointer	1000	0.06054	[-0.0765, 0.198]	0.7146	
vanilla	pointer	1e+04	0.05421	[-0.0828, 0.191]	0.7995	
vanilla	seq2seq	0.01	-0.002683	[-0.0348, 0.0294]	1.0000	
vanilla	seq2seq	0.1	-5.058×10^{-4}	[-0.0326, 0.0316]	1.0000	
vanilla	seq2seq	1	-0.002484	[-0.0346, 0.0296]	1.0000	
vanilla	seq2seq	10	0.01563	[-0.0164, 0.0477]	0.6259	
vanilla	seq2seq	100	0.01502	[-0.017, 0.0471]	0.6629	
vanilla	seq2seq	1000	-0.02409	[-0.0562, 0.00797]	0.2072	
vanilla	seq2seq	1e+04	-0.04363	[-0.0757, -0.0116]	0.0042	*

Table 3: `ood_len_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-0.004198	[-0.0112, 0.00103]	0.4432	-0.116	0.3915
adaptive	seq2seq	-0.00688	[-0.00807, -0.00571]	$< 10^{-4}$	-0.455	0.0004
tropical	pointer	0.001751	[-0.00287, 0.00605]	0.7314	0.0286	0.8382
tropical	seq2seq	-0.02888	[-0.0474, -0.0158]	0.0013	-0.429	0.0012
vanilla	pointer	-0.001912	[-0.0084, 0.00314]	0.7684	-0.0743	0.5889
vanilla	seq2seq	-0.005446	[-0.00743, -0.00301]	0.0117	-0.326	0.0136

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.08934	[0.00597, 0.173]	0.4093	
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.04602	[-0.0763, 0.168]	1.0000	
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.004687	[-0.122, 0.131]	1.0000	
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.07121	[-0.0835, 0.226]	1.0000	
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.04587	[-0.124, 0.215]	1.0000	
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.08953	[-0.0535, 0.233]	1.0000	
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.05592	[-0.121, 0.233]	1.0000	
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.03995	[-0.064, 0.144]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.2309	[0.132, 0.33]	0.0463	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.1565	[0.0256, 0.287]	0.3466	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.1912	[0.0322, 0.35]	0.3466	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.1868	[0.0697, 0.304]	0.1714	

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.1816	[0.0677, 0.296]	0.1714	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.08655	[0.0296, 0.144]	0.1753	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.1416	$[7.96 \times 10^{-5}, 0.283]$	0.4493	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.004291	[-0.0189, 0.0275]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.08913	[-0.0738, 0.252]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.01498	[-0.0695, 0.0995]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.02092	[-0.0638, 0.106]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.02144	[-0.0708, 0.114]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.02386	[-0.0718, 0.12]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.02943	[-0.0681, 0.127]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.005788	[-0.0753, 0.0869]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.001073	[-0.0905, 0.0926]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.003601	[-0.012, 0.00479]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	8.320×10^{-4}	[-0.0114, 0.013]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.002757	[-0.019, 0.0135]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.001145	[-0.0122, 0.0099]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.01361	[-0.0432, 0.016]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.0157	[-0.0363, 0.0049]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.007917	[-0.0209, 0.00511]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.004252	[-0.0129, 0.00439]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.2683	[0.159, 0.378]	0.0335	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.2607	[0.159, 0.363]	0.0335	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.2626	[0.162, 0.363]	0.0329	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.2653	[0.179, 0.351]	0.0204	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.2618	[0.158, 0.365]	0.0335	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.2638	[0.167, 0.361]	0.0295	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.2739	[0.182, 0.365]	0.0218	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.2787	[0.203, 0.354]	0.0123	*

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.03394	[-0.145, 0.213]	0.7623	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.1361	[0.0247, 0.247]	0.1647	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.0524	[-0.0956, 0.2]	0.7623	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.1271	[-0.0312, 0.285]	0.2691	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.08857	[0.0035, 0.174]	0.1781	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.2217	[0.0919, 0.351]	0.0631	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.1745	[0.0252, 0.324]	0.1647	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.309	[0.221, 0.397]	0.0134	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.1755	[0.121, 0.23]	0.0183	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.2466	[0.132, 0.361]	0.0396	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.2389	[0.124, 0.354]	0.0408	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.2427	[0.125, 0.36]	0.0408	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.2243	[0.134, 0.315]	0.0335	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.2187	[0.13, 0.307]	0.0335	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.2602	[0.156, 0.364]	0.0335	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.2734	[0.198, 0.349]	0.0125	*

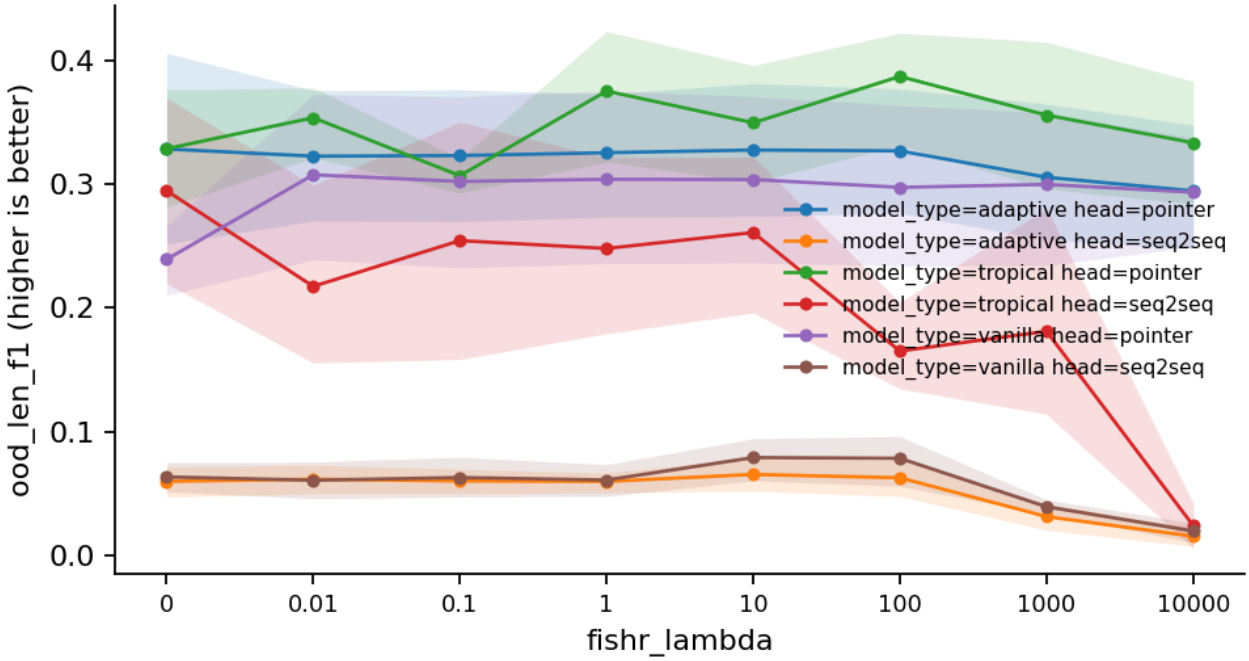


Figure 1: `ood_len_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

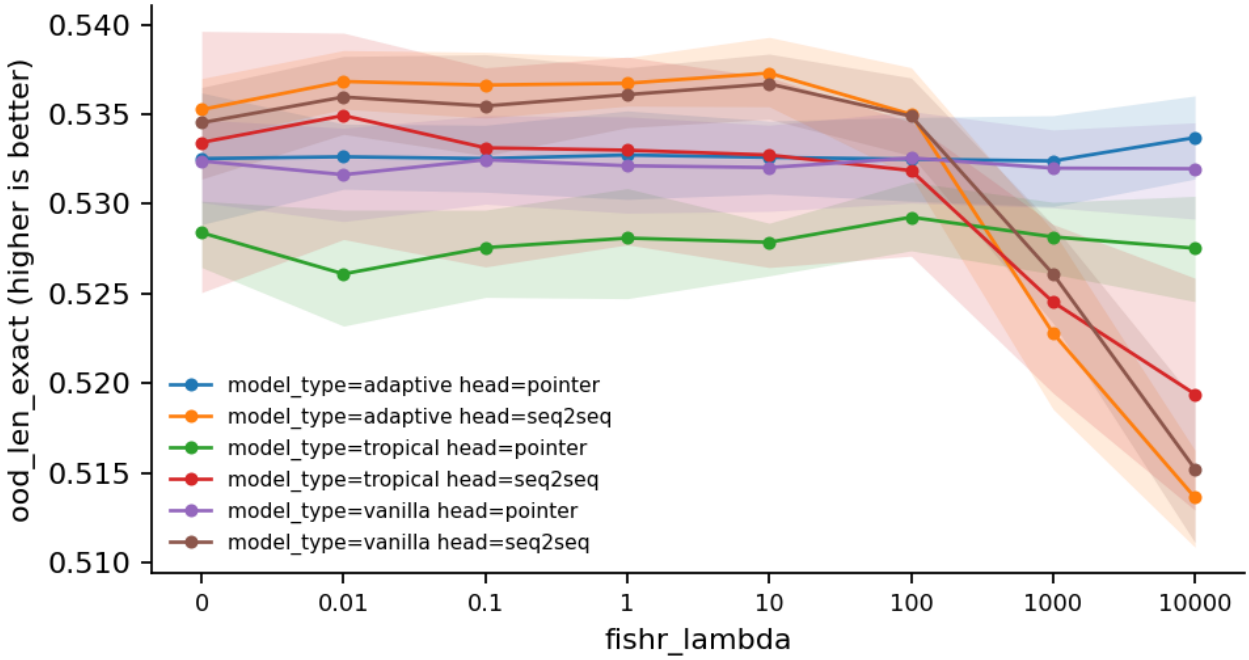


Figure 2: `ood_len_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A Other metrics

A.1 `ood_len_exact`

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.5325	[0.529, 0.536]	0.0048	5	0
adaptive	pointer	0.01	0.5326	[0.531, 0.534]	0.0024	5	0
adaptive	pointer	0.1	0.5325	[0.531, 0.534]	0.00249	5	0
adaptive	pointer	1	0.5327	[0.53, 0.535]	0.0032	5	0
adaptive	pointer	10	0.5326	[0.531, 0.535]	0.00259	5	0
adaptive	pointer	100	0.5325	[0.53, 0.535]	0.00297	5	0
adaptive	pointer	1000	0.5324	[0.53, 0.535]	0.00344	5	0
adaptive	pointer	1e+04	0.5337	[0.531, 0.536]	0.003	5	0
adaptive	seq2seq	0	0.5353	[0.534, 0.537]	0.00219	5	0
adaptive	seq2seq	0.01	0.5368	[0.535, 0.539]	0.0021	5	0
adaptive	seq2seq	0.1	0.5366	[0.535, 0.538]	0.00255	5	0
adaptive	seq2seq	1	0.5367	[0.535, 0.538]	0.00171	5	0
adaptive	seq2seq	10	0.5373	[0.535, 0.539]	0.00247	5	0
adaptive	seq2seq	100	0.535	[0.532, 0.538]	0.00414	5	0
adaptive	seq2seq	1000	0.5228	[0.519, 0.528]	0.00641	5	0
adaptive	seq2seq	1e+04	0.5136	[0.511, 0.516]	0.00366	5	0
tropical	pointer	0	0.5284	[0.526, 0.53]	0.0025	5	0
tropical	pointer	0.01	0.5261	[0.523, 0.53]	0.00403	5	0
tropical	pointer	0.1	0.5276	[0.525, 0.53]	0.00307	5	0
tropical	pointer	1	0.5281	[0.525, 0.531]	0.00398	5	0
tropical	pointer	10	0.5279	[0.526, 0.529]	0.00208	5	0
tropical	pointer	100	0.5293	[0.527, 0.531]	0.00241	5	0

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	1000	0.5282	[0.526, 0.53]	0.00258	5	0
tropical	pointer	1e+04	0.5275	[0.525, 0.53]	0.00382	5	0
tropical	seq2seq	0	0.5334	[0.525, 0.54]	0.00914	5	0
tropical	seq2seq	0.01	0.5349	[0.528, 0.54]	0.00779	5	0
tropical	seq2seq	0.1	0.5331	[0.527, 0.538]	0.0075	5	0
tropical	seq2seq	1	0.533	[0.528, 0.538]	0.0064	5	0
tropical	seq2seq	10	0.5327	[0.526, 0.537]	0.00751	5	0
tropical	seq2seq	100	0.5319	[0.527, 0.535]	0.00526	5	0
tropical	seq2seq	1000	0.5245	[0.519, 0.529]	0.00604	5	0
tropical	seq2seq	1e+04	0.5194	[0.513, 0.526]	0.00881	5	0
vanilla	pointer	0	0.5324	[0.53, 0.535]	0.00299	5	0
vanilla	pointer	0.01	0.5316	[0.529, 0.534]	0.00331	5	0
vanilla	pointer	0.1	0.5325	[0.53, 0.535]	0.00313	5	0
vanilla	pointer	1	0.5321	[0.529, 0.535]	0.00349	5	0
vanilla	pointer	10	0.532	[0.53, 0.534]	0.00305	5	0
vanilla	pointer	100	0.5326	[0.53, 0.535]	0.00335	5	0
vanilla	pointer	1000	0.532	[0.53, 0.534]	0.00271	5	0
vanilla	pointer	1e+04	0.532	[0.529, 0.535]	0.00344	5	0
vanilla	seq2seq	0	0.5345	[0.531, 0.536]	0.00354	5	0
vanilla	seq2seq	0.01	0.536	[0.534, 0.538]	0.00261	5	0
vanilla	seq2seq	0.1	0.5355	[0.533, 0.538]	0.0036	5	0
vanilla	seq2seq	1	0.5361	[0.534, 0.538]	0.00214	5	0
vanilla	seq2seq	10	0.5367	[0.535, 0.538]	0.00222	5	0
vanilla	seq2seq	100	0.5349	[0.533, 0.537]	0.00285	5	0
vanilla	seq2seq	1000	0.5261	[0.523, 0.529]	0.00365	5	0
vanilla	seq2seq	1e+04	0.5152	[0.511, 0.519]	0.00529	5	0

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	1.000×10^{-4}	[-0.00548, 0.00568]	1.0000	
adaptive	pointer	0.1	-1.110×10^{-16}	[-0.00558, 0.00558]	1.0000	
adaptive	pointer	1	2.000×10^{-4}	[-0.00538, 0.00578]	1.0000	
adaptive	pointer	10	6.667×10^{-5}	[-0.00551, 0.00565]	1.0000	
adaptive	pointer	100	-3.333×10^{-5}	[-0.00561, 0.00555]	1.0000	
adaptive	pointer	1000	-1.333×10^{-4}	[-0.00571, 0.00545]	1.0000	
adaptive	pointer	1e+04	0.001167	[-0.00441, 0.00675]	0.9894	
adaptive	seq2seq	0.01	0.001567	[-0.0045, 0.00763]	0.9673	
adaptive	seq2seq	0.1	0.001367	[-0.0047, 0.00743]	0.9839	
adaptive	seq2seq	1	0.001467	[-0.0046, 0.00753]	0.9766	
adaptive	seq2seq	10	0.002033	[-0.00403, 0.0081]	0.8923	
adaptive	seq2seq	100	-2.667×10^{-4}	[-0.00633, 0.0058]	1.0000	
adaptive	seq2seq	1000	-0.01247	[-0.0185, -0.0064]	$< 10^{-4}$	*
adaptive	seq2seq	1e+04	-0.02163	[-0.0277, -0.0156]	$< 10^{-4}$	*
tropical	pointer	0.01	-0.0023	[-0.00779, 0.00319]	0.7578	
tropical	pointer	0.1	-8.333×10^{-4}	[-0.00632, 0.00466]	0.9984	
tropical	pointer	1	-3.000×10^{-4}	[-0.00579, 0.00519]	1.0000	
tropical	pointer	10	-5.333×10^{-4}	[-0.00602, 0.00496]	0.9999	
tropical	pointer	100	8.667×10^{-4}	[-0.00462, 0.00636]	0.9980	
tropical	pointer	1000	-2.333×10^{-4}	[-0.00572, 0.00526]	1.0000	
tropical	pointer	1e+04	-8.667×10^{-4}	[-0.00636, 0.00462]	0.9980	
tropical	seq2seq	0.01	0.0015	[-0.0114, 0.0144]	0.9997	
tropical	seq2seq	0.1	-3.000×10^{-4}	[-0.0132, 0.0126]	1.0000	

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p	
tropical	seq2seq	1	-4.333×10^{-4}	[-0.0134, 0.0125]	1.0000	
tropical	seq2seq	10	-7.000×10^{-4}	[-0.0136, 0.0122]	1.0000	
tropical	seq2seq	100	-0.001567	[-0.0145, 0.0114]	0.9996	
tropical	seq2seq	1000	-0.0089	[-0.0218, 0.00405]	0.2833	
tropical	seq2seq	1e+04	-0.01403	[-0.027, -0.00108]	0.0292	*
vanilla	pointer	0.01	-7.667×10^{-4}	[-0.00635, 0.00481]	0.9991	
vanilla	pointer	0.1	6.667×10^{-5}	[-0.00551, 0.00565]	1.0000	
vanilla	pointer	1	-2.667×10^{-4}	[-0.00585, 0.00531]	1.0000	
vanilla	pointer	10	-3.667×10^{-4}	[-0.00595, 0.00521]	1.0000	
vanilla	pointer	100	1.667×10^{-4}	[-0.00541, 0.00575]	1.0000	
vanilla	pointer	1000	-4.000×10^{-4}	[-0.00598, 0.00518]	1.0000	
vanilla	pointer	1e+04	-4.333×10^{-4}	[-0.00601, 0.00515]	1.0000	
vanilla	seq2seq	0.01	0.001433	[-0.00446, 0.00733]	0.9761	
vanilla	seq2seq	0.1	9.333×10^{-4}	[-0.00496, 0.00683]	0.9979	
vanilla	seq2seq	1	0.001567	[-0.00433, 0.00746]	0.9625	
vanilla	seq2seq	10	0.002167	[-0.00373, 0.00806]	0.8463	
vanilla	seq2seq	100	3.667×10^{-4}	[-0.00553, 0.00626]	1.0000	
vanilla	seq2seq	1000	-0.008433	[-0.0143, -0.00254]	0.0025	*
vanilla	seq2seq	1e+04	-0.0193	[-0.0252, -0.0134]	$< 10^{-4}$	*

Table 7: `ood_len_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	9.643×10^{-5}	$[-7.50 \times 10^{-5}, 2.63 \times 10^{-4}]$	0.6743	0.0724	0.6001
adaptive	seq2seq	-0.003536	[-0.00406, -0.00286]	$< 10^{-4}$	-0.52	0.0002
tropical	pointer	2.381×10^{-4}	$[-2.26 \times 10^{-5}, 4.99 \times 10^{-4}]$	0.3649	0.105	0.4441
tropical	seq2seq	-0.002319	[-0.00357, -0.00107]	0.0005	-0.503	0.0004
vanilla	pointer	1.786×10^{-5}	$[-8.10 \times 10^{-5}, 1.17 \times 10^{-4}]$	0.9440	0.0552	0.6783
vanilla	seq2seq	-0.002933	[-0.00356, -0.00224]	$< 10^{-4}$	-0.497	0.0002

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	-0.004	[-0.00639, -0.00161]	0.1358	
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-0.005533	[-0.00939, -0.00168]	0.1987	
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.0049	[-0.00857, -0.00123]	0.2087	
model_type: tropical - vanilla head=pointer fishr_lambda=1	-0.004033	[-0.00679, -0.00128]	0.1987	
model_type: tropical - vanilla head=pointer fishr_lambda=10	-0.004167	[-0.0072, -0.00113]	0.2087	
model_type: tropical - vanilla head=pointer fishr_lambda=100	-0.0033	[-0.00468, -0.00192]	0.0432	*
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.003833	[-0.00561, -0.00205]	0.0590	
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-0.004433	$[-0.00851, -3.53 \times 10^{-4}]$	0.3537	

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.0011	[-0.00896, 0.00676]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.001033	[-0.00832, 0.00625]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.002333	[-0.00981, 0.00514]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.0031	[-0.00873, 0.00253]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.003967	[-0.0116, 0.00367]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.003033	[-0.00793, 0.00186]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.001567	[-0.00673, 0.0036]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.004167	[-0.00613, 0.0145]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	1.333×10^{-4}	[-0.00318, 0.00345]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.001	$[-5.21 \times 10^{-4}$, 0.00252]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	6.667×10^{-5}	[-0.00134, 0.00147]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	6.000×10^{-4}	$[1.24 \times 10^{-4}$, 0.00108]	0.3996	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	5.667×10^{-4}	$[-2.56 \times 10^{-4}$, 0.00139]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-6.667×10^{-5}	$[-7.62 \times 10^{-4}, 6.29 \times 10^{-4}]$	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	4.000×10^{-4}	$[-9.14 \times 10^{-4}$, 0.00171]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.001733	[-0.00167, 0.00513]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	7.333×10^{-4}	[-0.00272, 0.00419]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	8.667×10^{-4}	[-0.00245, 0.00418]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.001167	[-0.00234, 0.00467]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	6.333×10^{-4}	[-0.00166, 0.00293]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	6.000×10^{-4}	$[-7.14 \times 10^{-4}$, 0.00191]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	1.000×10^{-4}	[-0.00329, 0.00349]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.0033	[-0.00901, 0.00241]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.0016	[-0.00463, 0.00143]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.002733	[-0.00701, 0.00154]	0.7518	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.0042	[-0.00516, -0.00324]	0.0064	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.0041	[-0.00546, -0.00274]	0.0235	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.004	[-0.00626, -0.00174]	0.1467	

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.0047	[-0.00606, -0.00334]	0.0152	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.0025	[-0.00538, 3.79×10^{-4}]	0.7346	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.0096	[2.20×10^{-4}, 0.019]	0.6082	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.02007	[0.0139, 0.0263]	0.0188	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.005033	[-0.0141, 0.00404]	0.7606	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.008833	[-0.0176, -2.30×10^{-5}]	0.6082	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.005567	[-0.0122, 0.00104]	0.7346	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.0049	[-0.0106, 8.44×10^{-4}]	0.7346	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.004867	[-0.0118, 0.00208]	0.7407	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.0026	[-0.00799, 0.00279]	0.7606	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.003633	[-0.00277, 0.01]	0.7606	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.008133	[-0.00197, 0.0182]	0.7346	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.002133	[-0.00598, 0.00171]	0.7606	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.004333	[-0.00678, -0.00189]	0.1467	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.003	[-0.00609, 8.69×10^{-5}]	0.6082	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.003967	[-0.00669, -0.00124]	0.2345	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.004667	[-0.00728, -0.00205]	0.1467	
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.002333	[-0.00435, -3.16×10^{-4}]	0.4555	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.0059	[0.00249, 0.00931]	0.1467	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.01673	[0.0105, 0.023]	0.0346	*

A.2 `ood_val_f1`

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.2702	[0.257, 0.285]	0.0182	5	0
adaptive	pointer	0.01	0.2695	[0.251, 0.285]	0.0218	5	0
adaptive	pointer	0.1	0.268	[0.251, 0.285]	0.0217	5	0
adaptive	pointer	1	0.2693	[0.244, 0.29]	0.0293	5	0
adaptive	pointer	10	0.2742	[0.25, 0.295]	0.0291	5	0
adaptive	pointer	100	0.2691	[0.251, 0.288]	0.0238	5	0
adaptive	pointer	1000	0.272	[0.255, 0.289]	0.0212	5	0
adaptive	pointer	1e+04	0.2598	[0.235, 0.281]	0.0287	5	0

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	seq2seq	0	0.1438	[0.0846, 0.181]	0.0655	5	0
adaptive	seq2seq	0.01	0.1509	[0.0878, 0.208]	0.0766	5	0
adaptive	seq2seq	0.1	0.1466	[0.0906, 0.198]	0.0689	5	0
adaptive	seq2seq	1	0.1382	[0.076, 0.193]	0.0746	5	0
adaptive	seq2seq	10	0.1842	[0.134, 0.219]	0.0561	5	0
adaptive	seq2seq	100	0.1538	[0.115, 0.197]	0.0538	5	0
adaptive	seq2seq	1000	0.05334	[0.018, 0.0913]	0.0494	5	0
adaptive	seq2seq	1e+04	0.1349	[0.103, 0.167]	0.0405	5	0
tropical	pointer	0	0.301	[0.292, 0.31]	0.0113	5	0
tropical	pointer	0.01	0.3149	[0.292, 0.334]	0.0264	5	0
tropical	pointer	0.1	0.3279	[0.31, 0.342]	0.0217	5	0
tropical	pointer	1	0.3266	[0.317, 0.338]	0.0138	5	0
tropical	pointer	10	0.3362	[0.331, 0.342]	0.00739	5	0
tropical	pointer	100	0.3094	[0.281, 0.335]	0.0379	5	0
tropical	pointer	1000	0.3228	[0.316, 0.329]	0.00862	5	0
tropical	pointer	1e+04	0.3022	[0.288, 0.313]	0.0164	5	0
tropical	seq2seq	0	0.2825	[0.187, 0.363]	0.112	5	0
tropical	seq2seq	0.01	0.3282	[0.28, 0.376]	0.0629	5	0
tropical	seq2seq	0.1	0.2971	[0.219, 0.375]	0.1	5	0
tropical	seq2seq	1	0.3091	[0.24, 0.374]	0.0846	5	0
tropical	seq2seq	10	0.3408	[0.269, 0.397]	0.0839	5	0
tropical	seq2seq	100	0.2932	[0.23, 0.338]	0.0747	5	0
tropical	seq2seq	1000	0.315	[0.243, 0.374]	0.0822	5	0
tropical	seq2seq	1e+04	0.108	[0.0233, 0.202]	0.116	5	0
vanilla	pointer	0	0.2508	[0.235, 0.27]	0.0228	5	0
vanilla	pointer	0.01	0.2656	[0.247, 0.281]	0.0208	5	0
vanilla	pointer	0.1	0.2662	[0.248, 0.283]	0.0221	5	0
vanilla	pointer	1	0.265	[0.249, 0.279]	0.0197	5	0
vanilla	pointer	10	0.2658	[0.249, 0.279]	0.0189	5	0
vanilla	pointer	100	0.2659	[0.246, 0.283]	0.0234	5	0
vanilla	pointer	1000	0.2661	[0.25, 0.28]	0.0193	5	0
vanilla	pointer	1e+04	0.2611	[0.245, 0.277]	0.0213	5	0
vanilla	seq2seq	0	0.1952	[0.144, 0.243]	0.0651	5	0
vanilla	seq2seq	0.01	0.1789	[0.1, 0.25]	0.103	5	0
vanilla	seq2seq	0.1	0.19	[0.0993, 0.255]	0.103	5	0
vanilla	seq2seq	1	0.1702	[0.0982, 0.234]	0.0877	5	0
vanilla	seq2seq	10	0.1735	[0.119, 0.236]	0.0734	5	0
vanilla	seq2seq	100	0.1318	[0.0682, 0.194]	0.0795	5	0
vanilla	seq2seq	1000	0.05965	[0.0289, 0.109]	0.0536	5	0
vanilla	seq2seq	1e+04	0.158	[0.135, 0.181]	0.0287	5	0

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-7.575×10^{-4}	[-0.0437, 0.0422]	1.0000
adaptive	pointer	0.1	-0.00227	[-0.0452, 0.0406]	1.0000
adaptive	pointer	1	-9.359×10^{-4}	[-0.0439, 0.042]	1.0000
adaptive	pointer	10	0.00391	[-0.039, 0.0468]	0.9999
adaptive	pointer	100	-0.001113	[-0.044, 0.0418]	1.0000
adaptive	pointer	1000	0.001751	[-0.0412, 0.0447]	1.0000
adaptive	pointer	1e+04	-0.01047	[-0.0534, 0.0324]	0.9756
adaptive	seq2seq	0.01	0.007075	[-0.101, 0.115]	1.0000
adaptive	seq2seq	0.1	0.002804	[-0.105, 0.111]	1.0000

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	seq2seq	1	-0.005578	[-0.114, 0.102]	1.0000	
adaptive	seq2seq	10	0.04037	[-0.0676, 0.148]	0.8362	
adaptive	seq2seq	100	0.00994	[-0.0981, 0.118]	0.9999	
adaptive	seq2seq	1000	-0.09048	[-0.198, 0.0175]	0.1308	
adaptive	seq2seq	1e+04	-0.008894	[-0.117, 0.0991]	1.0000	
tropical	pointer	0.01	0.01381	[-0.0218, 0.0494]	0.8123	
tropical	pointer	0.1	0.0269	[-0.00869, 0.0625]	0.2023	
tropical	pointer	1	0.02554	[-0.01, 0.0611]	0.2450	
tropical	pointer	10	0.03519	$[-3.97 \times 10^{-4}$, 0.0708]	0.0537	
tropical	pointer	100	0.008386	[-0.0272, 0.044]	0.9796	
tropical	pointer	1000	0.02173	[-0.0139, 0.0573]	0.3982	
tropical	pointer	1e+04	0.001163	[-0.0344, 0.0368]	1.0000	
tropical	seq2seq	0.01	0.04574	[-0.114, 0.205]	0.9456	
tropical	seq2seq	0.1	0.01462	[-0.145, 0.174]	0.9999	
tropical	seq2seq	1	0.0266	[-0.133, 0.186]	0.9971	
tropical	seq2seq	10	0.05831	[-0.101, 0.218]	0.8485	
tropical	seq2seq	100	0.01078	[-0.149, 0.17]	1.0000	
tropical	seq2seq	1000	0.03254	[-0.127, 0.192]	0.9907	
tropical	seq2seq	1e+04	-0.1745	[-0.334, -0.0152]	0.0271	*
vanilla	pointer	0.01	0.01484	[-0.022, 0.0517]	0.7872	
vanilla	pointer	0.1	0.01538	[-0.0215, 0.0522]	0.7609	
vanilla	pointer	1	0.01415	[-0.0227, 0.051]	0.8195	
vanilla	pointer	10	0.01503	[-0.0218, 0.0519]	0.7780	
vanilla	pointer	100	0.01507	[-0.0218, 0.0519]	0.7761	
vanilla	pointer	1000	0.01524	[-0.0216, 0.0521]	0.7675	
vanilla	pointer	1e+04	0.01033	[-0.0265, 0.0472]	0.9514	
vanilla	seq2seq	0.01	-0.01634	[-0.152, 0.12]	0.9996	
vanilla	seq2seq	0.1	-0.005171	[-0.141, 0.131]	1.0000	
vanilla	seq2seq	1	-0.02504	[-0.161, 0.111]	0.9949	
vanilla	seq2seq	10	-0.02168	[-0.158, 0.114]	0.9978	
vanilla	seq2seq	100	-0.06344	[-0.199, 0.0726]	0.6671	
vanilla	seq2seq	1000	-0.1356	[-0.272, 4.57×10^{-4}]	0.0510	
vanilla	seq2seq	1e+04	-0.03718	[-0.173, 0.0989]	0.9569	

Table 11: `ood_val_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-7.598×10^{-4}	[-0.00131, -2.05×10^{-4}]	0.7068	-0.0171	0.9054
adaptive	seq2seq	-0.00782	[-0.0189, 0.00525]	0.1729	-0.192	0.1566
tropical	pointer	-0.002338	[-0.00405, -1.58×10^{-4}]	0.2203	-0.238	0.0746
tropical	seq2seq	-0.02288	[-0.0348, -0.0128]	0.0109	-0.265	0.0450
vanilla	pointer	-4.600×10^{-4}	[-0.0012, 5.42×10^{-4}]	0.7792	-0.0286	0.8432
vanilla	seq2seq	-0.01292	[-0.0241, 0.00418]	0.0661	-0.238	0.0736

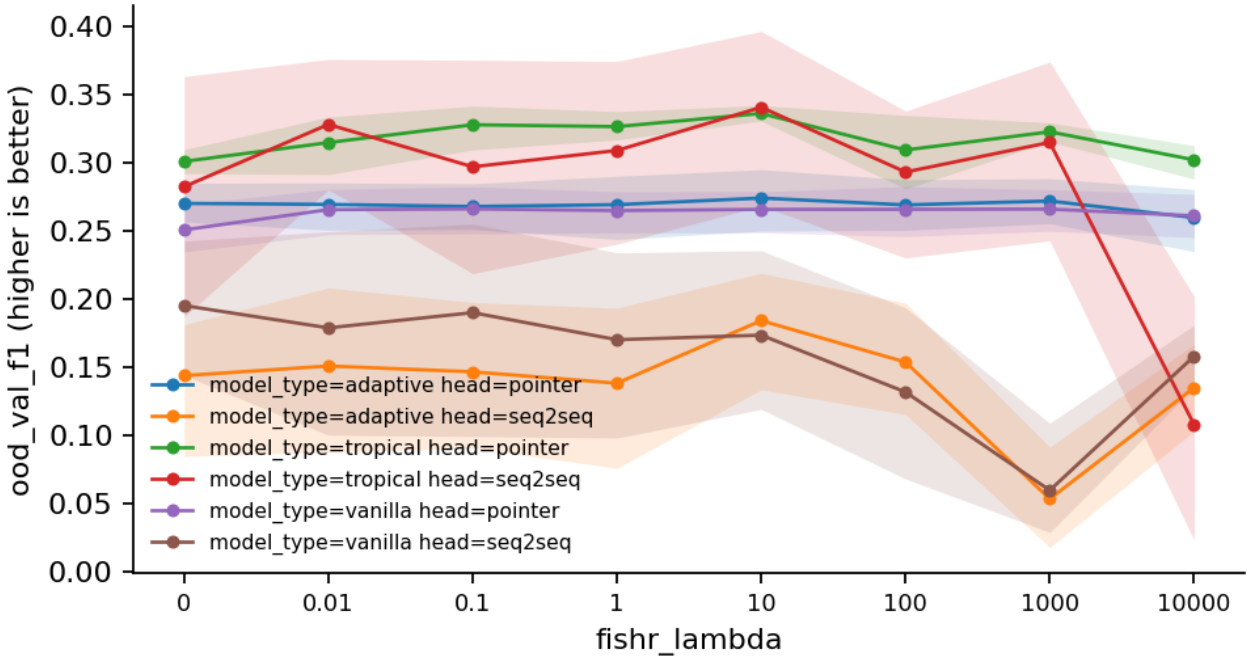


Figure 3: `ood_val_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.05023	[0.0216, 0.0788]	0.0819	
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.04921	[0.0211, 0.0773]	0.0819	
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.06176	[0.0226, 0.101]	0.0829	
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.06163	[0.0333, 0.0899]	0.0454	*
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.0704	[0.0471, 0.0937]	0.0166	*
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.04355	[0.00532, 0.0818]	0.1705	
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.05672	[0.0319, 0.0815]	0.0441	*
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.04107	[0.00273, 0.0794]	0.1705	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.08725	[-0.0249, 0.199]	0.1937	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.1493	[0.0606, 0.238]	0.0819	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.107	[-0.00631, 0.22]	0.1761	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.1389	[0.0486, 0.229]	0.0829	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.1672	[0.0832, 0.251]	0.0577	

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.1615	[0.109, 0.214]	0.0162	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.2554	[0.143, 0.368]	0.0441	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.05008	[-0.211, 0.111]	0.4355	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.01943	[-0.00815, 0.047]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.003834	[-0.0386, 0.0463]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.001782	[-0.0421, 0.0457]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.004345	[-0.0465, 0.0552]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.008312	[-0.0443, 0.0609]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.003252	[-0.0424, 0.0489]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.005938	[-0.034, 0.0459]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.001369	[-0.056, 0.0533]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.0514	[-0.114, 0.0115]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.02798	[-0.124, 0.0679]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.04342	[-0.135, 0.0483]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.03194	[-0.0932, 0.0293]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.01065	[-0.0511, 0.0724]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.02198	[-0.0579, 0.102]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.006307	[-0.0316, 0.019]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.02312	[-0.0426, -0.00359]	0.4850	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.1264	[0.0461, 0.207]	0.2277	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.1186	[0.0311, 0.206]	0.2778	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.1213	[0.0395, 0.203]	0.2469	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.1311	[0.0505, 0.212]	0.2139	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.08997	[0.0449, 0.135]	0.1087	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.1154	[0.0378, 0.193]	0.2469	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.2187	[0.149, 0.288]	0.0217	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.1248	[0.0427, 0.207]	0.2422	
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.01858	[-0.122, 0.159]	1.0000	

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.01334	[-0.109, 0.082]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.03086	[-0.115, 0.177]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.01752	[-0.0874, 0.122]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.004535	[-0.109, 0.0998]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.01619	[-0.105, 0.138]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.007765	[-0.0968, 0.112]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.1943	[0.0537, 0.335]	0.2778	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.0556	[-0.0222, 0.133]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.08678	[-0.0478, 0.221]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.07614	[-0.0637, 0.216]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.09478	[-0.0319, 0.221]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.09231	[-0.0166, 0.201]	0.9382	
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.1341	[0.013, 0.255]	0.4826	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.2064	[0.12, 0.293]	0.0585	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.1031	[0.0794, 0.127]	0.0065	*

A.3 `ood_val_exact`

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.3467	[0.334, 0.365]	0.0209	5	0
adaptive	pointer	0.01	0.3441	[0.325, 0.36]	0.0231	5	0
adaptive	pointer	0.1	0.3487	[0.329, 0.369]	0.0281	5	0
adaptive	pointer	1	0.3477	[0.327, 0.37]	0.0287	5	0
adaptive	pointer	10	0.3424	[0.324, 0.363]	0.0245	5	0
adaptive	pointer	100	0.3477	[0.333, 0.363]	0.0202	5	0
adaptive	pointer	1000	0.3414	[0.32, 0.357]	0.0246	5	0
adaptive	pointer	1e+04	0.3625	[0.345, 0.384]	0.0257	5	0
adaptive	seq2seq	0	0.4971	[0.487, 0.507]	0.013	5	0
adaptive	seq2seq	0.01	0.4936	[0.482, 0.506]	0.015	5	0
adaptive	seq2seq	0.1	0.498	[0.494, 0.503]	0.00558	5	0
adaptive	seq2seq	1	0.4988	[0.489, 0.506]	0.0115	5	0
adaptive	seq2seq	10	0.4805	[0.443, 0.508]	0.0418	5	0
adaptive	seq2seq	100	0.4867	[0.446, 0.524]	0.0493	5	0
adaptive	seq2seq	1000	0.5144	[0.503, 0.525]	0.0141	5	0
adaptive	seq2seq	1e+04	0.4288	[0.403, 0.454]	0.0359	5	0
tropical	pointer	0	0.25	[0.221, 0.275]	0.0347	5	0
tropical	pointer	0.01	0.2443	[0.2, 0.289]	0.0587	5	0

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	0.1	0.2406	[0.213, 0.267]	0.0351	5	0
tropical	pointer	1	0.2384	[0.2, 0.279]	0.0522	5	0
tropical	pointer	10	0.2287	[0.189, 0.268]	0.0512	5	0
tropical	pointer	100	0.2609	[0.215, 0.307]	0.061	5	0
tropical	pointer	1000	0.2335	[0.206, 0.264]	0.037	5	0
tropical	pointer	1e+04	0.2591	[0.234, 0.289]	0.0354	5	0
tropical	seq2seq	0	0.2906	[0.204, 0.389]	0.119	5	0
tropical	seq2seq	0.01	0.2807	[0.224, 0.334]	0.0722	5	0
tropical	seq2seq	0.1	0.3053	[0.224, 0.381]	0.101	5	0
tropical	seq2seq	1	0.3073	[0.228, 0.382]	0.102	5	0
tropical	seq2seq	10	0.2455	[0.167, 0.332]	0.105	5	0
tropical	seq2seq	100	0.2882	[0.233, 0.356]	0.0793	5	0
tropical	seq2seq	1000	0.2569	[0.208, 0.308]	0.0649	5	0
tropical	seq2seq	1e+04	0.4553	[0.393, 0.505]	0.0735	5	0
vanilla	pointer	0	0.323	[0.26, 0.374]	0.0749	5	0
vanilla	pointer	0.01	0.3284	[0.28, 0.377]	0.0651	5	0
vanilla	pointer	0.1	0.3235	[0.269, 0.379]	0.07	5	0
vanilla	pointer	1	0.3306	[0.28, 0.381]	0.0672	5	0
vanilla	pointer	10	0.3278	[0.277, 0.381]	0.067	5	0
vanilla	pointer	100	0.326	[0.27, 0.382]	0.0724	5	0
vanilla	pointer	1000	0.3308	[0.279, 0.379]	0.0653	5	0
vanilla	pointer	1e+04	0.3275	[0.277, 0.376]	0.0666	5	0
vanilla	seq2seq	0	0.4629	[0.441, 0.485]	0.0288	5	0
vanilla	seq2seq	0.01	0.4731	[0.442, 0.504]	0.0405	5	0
vanilla	seq2seq	0.1	0.4675	[0.437, 0.498]	0.0407	5	0
vanilla	seq2seq	1	0.4744	[0.437, 0.513]	0.0468	5	0
vanilla	seq2seq	10	0.4489	[0.423, 0.473]	0.0325	5	0
vanilla	seq2seq	100	0.4846	[0.453, 0.516]	0.0425	5	0
vanilla	seq2seq	1000	0.5091	[0.496, 0.52]	0.016	5	0
vanilla	seq2seq	1e+04	0.4152	[0.397, 0.435]	0.0248	5	0

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	-0.00262	[-0.0457, 0.0404]	1.0000	
adaptive	pointer	0.1	0.00194	[-0.0411, 0.045]	1.0000	
adaptive	pointer	1	0.001	[-0.0421, 0.0441]	1.0000	
adaptive	pointer	10	-0.00438	[-0.0474, 0.0387]	0.9999	
adaptive	pointer	100	9.600×10^{-4}	[-0.0421, 0.044]	1.0000	
adaptive	pointer	1000	-0.00536	[-0.0484, 0.0377]	0.9995	
adaptive	pointer	1e+04	0.0158	[-0.0273, 0.0589]	0.8469	
adaptive	seq2seq	0.01	-0.00348	[-0.0522, 0.0452]	1.0000	
adaptive	seq2seq	0.1	8.400×10^{-4}	[-0.0479, 0.0496]	1.0000	
adaptive	seq2seq	1	0.0017	[-0.047, 0.0504]	1.0000	
adaptive	seq2seq	10	-0.0166	[-0.0653, 0.0321]	0.8853	
adaptive	seq2seq	100	-0.01046	[-0.0592, 0.0383]	0.9877	
adaptive	seq2seq	1000	0.01728	[-0.0314, 0.066]	0.8656	
adaptive	seq2seq	1e+04	-0.06832	[-0.117, -0.0196]	0.0031	*
tropical	pointer	0.01	-0.00572	[-0.0876, 0.0761]	1.0000	
tropical	pointer	0.1	-0.00944	[-0.0913, 0.0724]	0.9997	
tropical	pointer	1	-0.0116	[-0.0935, 0.0703]	0.9990	
tropical	pointer	10	-0.02136	[-0.103, 0.0605]	0.9656	
tropical	pointer	100	0.01084	[-0.071, 0.0927]	0.9993	

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p	
tropical	pointer	1000	-0.01654	[-0.0984, 0.0653]	0.9912	
tropical	pointer	1e+04	0.00906	[-0.0728, 0.0909]	0.9998	
tropical	seq2seq	0.01	-0.00986	[-0.17, 0.15]	1.0000	
tropical	seq2seq	0.1	0.01474	[-0.145, 0.175]	0.9999	
tropical	seq2seq	1	0.0167	[-0.143, 0.177]	0.9999	
tropical	seq2seq	10	-0.04508	[-0.205, 0.115]	0.9498	
tropical	seq2seq	100	-0.00238	[-0.162, 0.157]	1.0000	
tropical	seq2seq	1000	-0.03368	[-0.193, 0.126]	0.9889	
tropical	seq2seq	1e+04	0.1647	[0.0049, 0.325]	0.0413	*
vanilla	pointer	0.01	0.00536	[-0.115, 0.125]	1.0000	
vanilla	pointer	0.1	4.400×10^{-4}	[-0.119, 0.12]	1.0000	
vanilla	pointer	1	0.00758	[-0.112, 0.127]	1.0000	
vanilla	pointer	10	0.0048	[-0.115, 0.125]	1.0000	
vanilla	pointer	100	0.00294	[-0.117, 0.123]	1.0000	
vanilla	pointer	1000	0.00782	[-0.112, 0.128]	1.0000	
vanilla	pointer	1e+04	0.00452	[-0.115, 0.124]	1.0000	
vanilla	seq2seq	0.01	0.01018	[-0.0517, 0.0721]	0.9974	
vanilla	seq2seq	0.1	0.00456	[-0.0574, 0.0665]	1.0000	
vanilla	seq2seq	1	0.01152	[-0.0504, 0.0734]	0.9946	
vanilla	seq2seq	10	-0.014	[-0.0759, 0.0479]	0.9837	
vanilla	seq2seq	100	0.0217	[-0.0402, 0.0836]	0.8717	
vanilla	seq2seq	1000	0.0462	[-0.0157, 0.108]	0.2128	
vanilla	seq2seq	1e+04	-0.04768	[-0.11, 0.0142]	0.1883	

Table 15: `ood_val_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.001451	$[3.36 \times 10^{-4}$, 0.00215]	0.4783	0.099	0.4517
adaptive	seq2seq	-0.006207	[-0.00934, -0.00308]	0.0455	-0.124	0.3551
tropical	pointer	0.001878	[-0.00208, 0.00485]	0.6314	0.0476	0.7267
tropical	seq2seq	0.01456	[0.00587, 0.0283]	0.0934	0.177	0.1800
vanilla	pointer	2.714×10^{-4}	$[-8.26 \times 10^{-4}$, 0.00159]	0.9593	-0.00952	0.9516
vanilla	seq2seq	-0.002861	[-0.00871, 0.00305]	0.4391	-0.0914	0.4887

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	-0.07298	[-0.206, 0.0602]	0.5887
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-0.08406	[-0.186, 0.018]	0.5576
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.08286	[-0.168, 0.00239]	0.4875
model_type: tropical - vanilla head=pointer fishr_lambda=1	-0.09216	[-0.214, 0.0298]	0.5576
model_type: tropical - vanilla head=pointer fishr_lambda=10	-0.09914	[-0.216, 0.0174]	0.5576
model_type: tropical - vanilla head=pointer fishr_lambda=100	-0.06508	[-0.169, 0.0391]	0.5887

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.09734	[-0.207, 0.0125]	0.5576	
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-0.06844	[-0.174, 0.0374]	0.5887	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.1723	[-0.295, -0.0493]	0.1871	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.1923	[-0.251, -0.133]	0.0123	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.1621	[-0.271, -0.0534]	0.1722	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.1671	[-0.285, -0.0492]	0.1871	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.2034	[-0.327, -0.0799]	0.1330	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.1964	[-0.265, -0.127]	0.0194	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.2522	[-0.327, -0.178]	0.0115	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.04008	[-0.0672, 0.147]	0.5887	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.02372	[-0.0918, 0.139]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.01574	[-0.0832, 0.115]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.02522	[-0.0713, 0.122]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.01714	[-0.0923, 0.127]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.01454	[-0.0949, 0.124]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.02174	[-0.0741, 0.118]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.01054	[-0.0922, 0.113]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.035	[-0.0718, 0.142]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.03422	[-0.00104, 0.0695]	0.8702	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.02056	[-0.0323, 0.0734]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.0305	[-0.0256, 0.0866]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.0244	[-0.0352, 0.084]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.03162	[-0.0162, 0.0794]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.00206	[-0.0562, 0.0603]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.0053	[-0.00838, 0.019]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.01358	[-0.00377, 0.0309]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.1504	[-0.19, -0.111]	0.0090	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.1495	[-0.181, -0.118]	0.0042	*

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.1493	[-0.181, -0.118]	0.0043	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.1511	[-0.185, -0.117]	0.0051	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.1382	[-0.201, -0.0751]	0.0701	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.139	[-0.211, -0.0672]	0.0960	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.173	[-0.202, -0.144]	0.0019	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.06626	[-0.127, -0.00541]	0.3124	
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.04056	[-0.176, 0.0948]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.03642	[-0.163, 0.0904]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.06474	[-0.223, 0.0936]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.06886	[-0.187, 0.0497]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.01684	[-0.164, 0.131]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.02734	[-0.193, 0.138]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.02342	[-0.121, 0.0738]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.1962	[-0.297, -0.0956]	0.0960	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.1399	[-0.234, -0.0461]	0.2013	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.1447	[-0.253, -0.0366]	0.2465	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.144	[-0.259, -0.0293]	0.2773	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.1438	[-0.266, -0.0217]	0.2905	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.1211	[-0.222, -0.0202]	0.2905	
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.1586	[-0.26, -0.0576]	0.1809	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.1783	[-0.268, -0.0889]	0.0933	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.08768	[-0.15, -0.0258]	0.2221	

A.4 `ood_noise_f1`

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.2241	[0.186, 0.265]	0.0488	5	0
adaptive	pointer	0.01	0.2401	[0.22, 0.26]	0.0259	5	0
adaptive	pointer	0.1	0.242	[0.221, 0.263]	0.0271	5	0
adaptive	pointer	1	0.2395	[0.218, 0.265]	0.0295	5	0

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	10	0.2369	[0.21, 0.268]	0.0367	5	0
adaptive	pointer	100	0.2397	[0.217, 0.263]	0.03	5	0
adaptive	pointer	1000	0.2292	[0.197, 0.258]	0.0382	5	0
adaptive	pointer	1e+04	0.2312	[0.201, 0.266]	0.0422	5	0
adaptive	seq2seq	0	0.1409	[0.0916, 0.191]	0.0642	5	0
adaptive	seq2seq	0.01	0.1135	[0.0564, 0.171]	0.0727	5	0
adaptive	seq2seq	0.1	0.1093	[0.0482, 0.166]	0.0778	5	0
adaptive	seq2seq	1	0.1281	[0.0691, 0.187]	0.0761	5	0
adaptive	seq2seq	10	0.1606	[0.113, 0.215]	0.0652	5	0
adaptive	seq2seq	100	0.1299	[0.0849, 0.174]	0.0579	5	0
adaptive	seq2seq	1000	0.05033	[0.015, 0.103]	0.0616	5	0
adaptive	seq2seq	1e+04	0.05052	[0.00575, 0.111]	0.0686	5	0
tropical	pointer	0	0.2817	[0.262, 0.302]	0.0281	5	0
tropical	pointer	0.01	0.2944	[0.278, 0.308]	0.0198	5	0
tropical	pointer	0.1	0.289	[0.275, 0.301]	0.0157	5	0
tropical	pointer	1	0.291	[0.273, 0.306]	0.0217	5	0
tropical	pointer	10	0.286	[0.258, 0.306]	0.0315	5	0
tropical	pointer	100	0.2817	[0.261, 0.302]	0.0257	5	0
tropical	pointer	1000	0.2893	[0.278, 0.3]	0.0142	5	0
tropical	pointer	1e+04	0.2886	[0.263, 0.314]	0.0331	5	0
tropical	seq2seq	0	0.2089	[0.139, 0.271]	0.0844	5	0
tropical	seq2seq	0.01	0.1863	[0.118, 0.244]	0.0825	5	0
tropical	seq2seq	0.1	0.1679	[0.0941, 0.229]	0.0895	5	0
tropical	seq2seq	1	0.2122	[0.152, 0.281]	0.0806	5	0
tropical	seq2seq	10	0.2299	[0.17, 0.285]	0.0766	5	0
tropical	seq2seq	100	0.2406	[0.181, 0.284]	0.0683	5	0
tropical	seq2seq	1000	0.2442	[0.194, 0.292]	0.0662	5	0
tropical	seq2seq	1e+04	0.1276	[0.0389, 0.216]	0.12	5	0
vanilla	pointer	0	0.2521	[0.23, 0.274]	0.0299	5	0
vanilla	pointer	0.01	0.2342	[0.204, 0.261]	0.0379	5	0
vanilla	pointer	0.1	0.232	[0.205, 0.262]	0.0369	5	0
vanilla	pointer	1	0.237	[0.209, 0.261]	0.0343	5	0
vanilla	pointer	10	0.2385	[0.211, 0.26]	0.0329	5	0
vanilla	pointer	100	0.2349	[0.21, 0.259]	0.0331	5	0
vanilla	pointer	1000	0.2244	[0.194, 0.255]	0.0391	5	0
vanilla	pointer	1e+04	0.2408	[0.21, 0.27]	0.0372	5	0
vanilla	seq2seq	0	0.1357	[0.0684, 0.199]	0.0834	5	0
vanilla	seq2seq	0.01	0.121	[0.0669, 0.175]	0.0713	5	0
vanilla	seq2seq	0.1	0.1375	[0.0736, 0.19]	0.0761	5	0
vanilla	seq2seq	1	0.1158	[0.0598, 0.165]	0.0673	5	0
vanilla	seq2seq	10	0.1478	[0.0814, 0.207]	0.0844	5	0
vanilla	seq2seq	100	0.1364	[0.109, 0.163]	0.0353	5	0
vanilla	seq2seq	1000	0.09003	[0.0463, 0.134]	0.0558	5	0
vanilla	seq2seq	1e+04	0.09908	[0.0158, 0.182]	0.108	5	0

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0.01601	[-0.0462, 0.0782]	0.9680
adaptive	pointer	0.1	0.01792	[-0.0443, 0.0802]	0.9448
adaptive	pointer	1	0.01537	[-0.0469, 0.0776]	0.9740
adaptive	pointer	10	0.01279	[-0.0494, 0.075]	0.9904
adaptive	pointer	100	0.01555	[-0.0467, 0.0778]	0.9724

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	1000	0.005084	[-0.0571, 0.0673]	1.0000
adaptive	pointer	1e+04	0.007099	[-0.0551, 0.0693]	0.9997
adaptive	seq2seq	0.01	-0.02741	[-0.147, 0.092]	0.9823
adaptive	seq2seq	0.1	-0.03163	[-0.151, 0.0878]	0.9630
adaptive	seq2seq	1	-0.01289	[-0.132, 0.107]	0.9998
adaptive	seq2seq	10	0.01967	[-0.0997, 0.139]	0.9974
adaptive	seq2seq	100	-0.01104	[-0.13, 0.108]	0.9999
adaptive	seq2seq	1000	-0.09062	[-0.21, 0.0288]	0.1992
adaptive	seq2seq	1e+04	-0.09043	[-0.21, 0.029]	0.2009
tropical	pointer	0.01	0.01268	[-0.0304, 0.0557]	0.9386
tropical	pointer	0.1	0.00728	[-0.0358, 0.0503]	0.9969
tropical	pointer	1	0.009293	[-0.0337, 0.0523]	0.9873
tropical	pointer	10	0.004316	[-0.0387, 0.0474]	0.9999
tropical	pointer	100	3.042×10^{-5}	[-0.043, 0.0431]	1.0000
tropical	pointer	1000	0.007605	[-0.0354, 0.0506]	0.9960
tropical	pointer	1e+04	0.00688	[-0.0362, 0.0499]	0.9978
tropical	seq2seq	0.01	-0.02257	[-0.171, 0.126]	0.9984
tropical	seq2seq	0.1	-0.04094	[-0.189, 0.107]	0.9547
tropical	seq2seq	1	0.003288	[-0.145, 0.152]	1.0000
tropical	seq2seq	10	0.02106	[-0.127, 0.169]	0.9989
tropical	seq2seq	100	0.03172	[-0.117, 0.18]	0.9879
tropical	seq2seq	1000	0.0353	[-0.113, 0.184]	0.9785
tropical	seq2seq	1e+04	-0.08125	[-0.229, 0.067]	0.5088
vanilla	pointer	0.01	-0.01783	[-0.0794, 0.0438]	0.9434
vanilla	pointer	0.1	-0.02001	[-0.0816, 0.0416]	0.9059
vanilla	pointer	1	-0.01502	[-0.0766, 0.0466]	0.9757
vanilla	pointer	10	-0.01357	[-0.0752, 0.048]	0.9858
vanilla	pointer	100	-0.0172	[-0.0788, 0.0444]	0.9522
vanilla	pointer	1000	-0.02762	[-0.0892, 0.034]	0.7022
vanilla	pointer	1e+04	-0.01127	[-0.0729, 0.0503]	0.9951
vanilla	seq2seq	0.01	-0.01463	[-0.146, 0.117]	0.9998
vanilla	seq2seq	0.1	0.001793	[-0.13, 0.134]	1.0000
vanilla	seq2seq	1	-0.01987	[-0.152, 0.112]	0.9985
vanilla	seq2seq	10	0.01217	[-0.12, 0.144]	0.9999
vanilla	seq2seq	100	7.394×10^{-4}	[-0.131, 0.132]	1.0000
vanilla	seq2seq	1000	-0.04564	[-0.177, 0.0861]	0.8773
vanilla	seq2seq	1e+04	-0.03659	[-0.168, 0.0951]	0.9534

Table 19: `ood_noise_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-0.001865	[-0.00419, 5.55×10^{-4}]	0.4788	-0.0933	0.4925
adaptive	seq2seq	-0.0109	[-0.0217, -1.49×10^{-4}]	0.0796	-0.261	0.0574
tropical	pointer	-9.290×10^{-4}	[-0.00252, 2.23×10^{-4}]	0.6275	-0.0438	0.7546
tropical	seq2seq	1.739×10^{-4}	[-0.00987, 0.0102]	0.9816	0.0476	0.7327
vanilla	pointer	8.126×10^{-5}	[-0.00182, 0.00156]	0.9773	-0.0019	1.0000
vanilla	seq2seq	-0.005005	[-0.0129, 0.00293]	0.4071	-0.135	0.3073

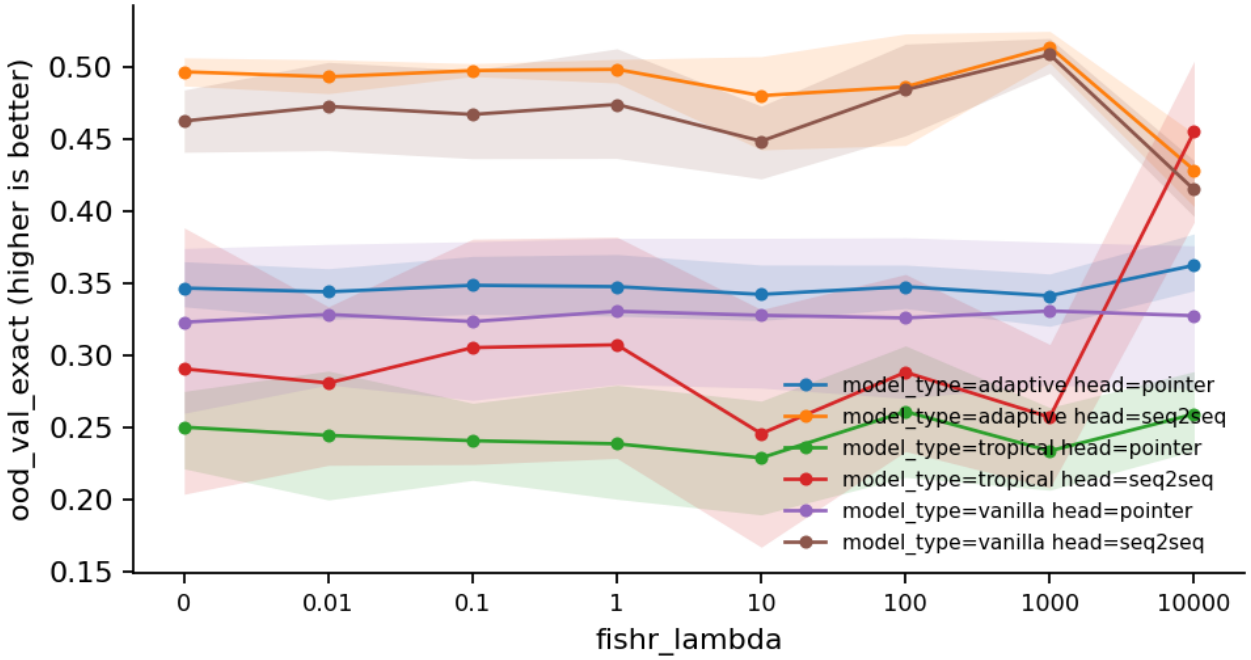


Figure 4: `ood_val_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

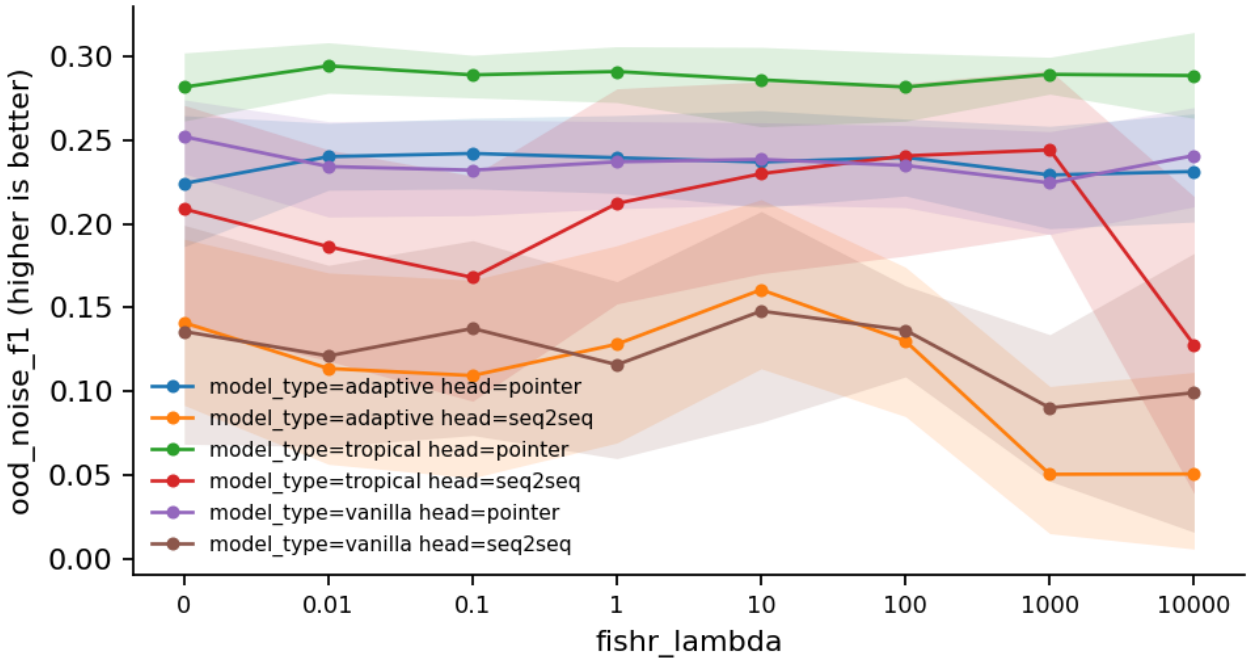


Figure 5: `ood_noise_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.02964	[-0.0188, 0.0781]	0.7839
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.06015	[0.0153, 0.105]	0.2848
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.05693	[0.0117, 0.102]	0.2848
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.05396	[0.00143, 0.106]	0.4631
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.04753	[-0.0122, 0.107]	0.6414
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.04687	[-0.00673, 0.1]	0.5772
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.06487	[0.015, 0.115]	0.2848
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.04779	[0.0227, 0.0729]	0.0979
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.0732	[-0.0529, 0.199]	0.7839
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.06526	[-0.034, 0.165]	0.7839
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.03047	[-0.0913, 0.152]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.09637	[-0.00795, 0.201]	0.5609
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.08209	[-0.038, 0.202]	0.7839
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.1042	[0.0252, 0.183]	0.2848
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.1541	[0.0457, 0.263]	0.2532
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.02855	[-0.112, 0.169]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.02793	[-0.112, 0.0559]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.005904	[-0.0446, 0.0564]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.009998	[-0.0373, 0.0573]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.002456	[-0.0495, 0.0544]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.001579	[-0.0461, 0.0429]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.004817	[-0.0405, 0.0501]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.004773	[-0.0451, 0.0547]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.009563	[-0.0542, 0.035]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.00528	[-0.0622, 0.0727]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.007503	[-0.117, 0.102]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.02815	[-0.129, 0.0729]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.01227	[-0.0992, 0.124]	1.0000

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.01277	[-0.0304, 0.0559]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.006504	[-0.101, 0.0884]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.03971	[-0.107, 0.0272]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.04856	[-0.11, 0.0131]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.08317	[-0.0079, 0.174]	0.4500	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.1266	[0.0306, 0.223]	0.3941	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.1327	[0.0274, 0.238]	0.4242	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.1114	[0.0114, 0.211]	0.4419	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.07629	[0.0137, 0.139]	0.4419	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.1098	[0.0127, 0.207]	0.4419	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.1789	[0.113, 0.245]	0.0390	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.1807	[0.0956, 0.266]	0.0910	
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.07282	[-0.0578, 0.203]	0.6654	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.1081	[0.00245, 0.214]	0.4419	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.121	[0.0155, 0.227]	0.4419	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.07883	[-0.0211, 0.179]	0.4678	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.05608	[-0.0515, 0.164]	0.6654	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.04113	[-0.0324, 0.115]	0.6654	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.04513	[-0.029, 0.119]	0.6654	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.161	[0.0403, 0.282]	0.3941	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.1164	[0.0333, 0.199]	0.3543	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.1132	[0.0185, 0.208]	0.4419	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.09458	[0.00615, 0.183]	0.4419	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.1212	[0.0203, 0.222]	0.4419	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.09064	[-0.0107, 0.192]	0.4500	
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.09844	[0.0485, 0.148]	0.1142	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.1344	[0.103, 0.165]	0.0065	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.1417	[0.0232, 0.26]	0.4419	

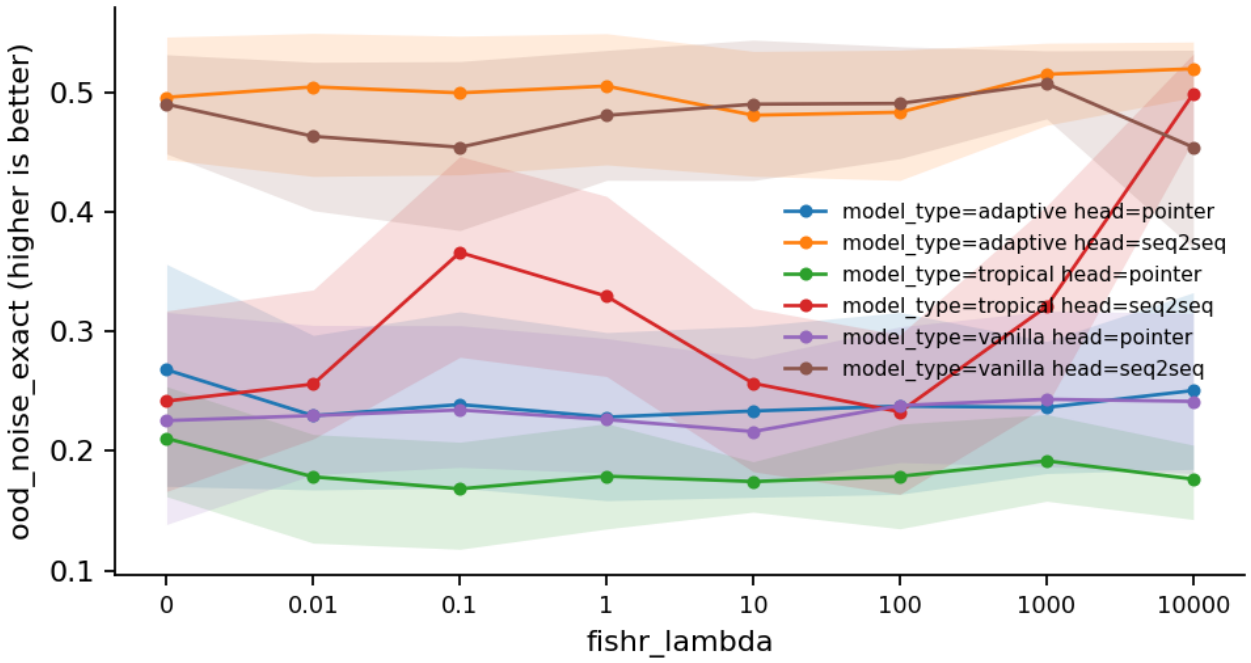


Figure 6: `ood_noise_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A.5 `ood_noise_exact`

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.268	[0.171, 0.356]	0.12	5	0
adaptive	pointer	0.01	0.2296	[0.168, 0.297]	0.0871	5	0
adaptive	pointer	0.1	0.2389	[0.169, 0.317]	0.0952	5	0
adaptive	pointer	1	0.2284	[0.159, 0.299]	0.0931	5	0
adaptive	pointer	10	0.2335	[0.161, 0.304]	0.0895	5	0
adaptive	pointer	100	0.2376	[0.164, 0.316]	0.0988	5	0
adaptive	pointer	1000	0.2365	[0.181, 0.292]	0.0749	5	0
adaptive	pointer	1e+04	0.2506	[0.185, 0.333]	0.0964	5	0
adaptive	seq2seq	0	0.4957	[0.444, 0.546]	0.0645	5	0
adaptive	seq2seq	0.01	0.5044	[0.43, 0.549]	0.0824	5	0
adaptive	seq2seq	0.1	0.4994	[0.431, 0.547]	0.0823	5	0
adaptive	seq2seq	1	0.5051	[0.439, 0.549]	0.073	5	0
adaptive	seq2seq	10	0.4808	[0.43, 0.534]	0.0683	5	0
adaptive	seq2seq	100	0.4832	[0.426, 0.535]	0.0764	5	0
adaptive	seq2seq	1000	0.5149	[0.472, 0.541]	0.0475	5	0
adaptive	seq2seq	1e+04	0.5195	[0.496, 0.542]	0.0298	5	0
tropical	pointer	0	0.2106	[0.162, 0.254]	0.0589	5	0
tropical	pointer	0.01	0.1785	[0.123, 0.214]	0.0577	5	0
tropical	pointer	0.1	0.1684	[0.118, 0.207]	0.0576	5	0
tropical	pointer	1	0.179	[0.135, 0.223]	0.0558	5	0
tropical	pointer	10	0.1745	[0.149, 0.191]	0.0286	5	0
tropical	pointer	100	0.1789	[0.135, 0.223]	0.0566	5	0
tropical	pointer	1000	0.1919	[0.158, 0.231]	0.048	5	0
tropical	pointer	1e+04	0.1766	[0.143, 0.205]	0.0404	5	0
tropical	seq2seq	0	0.2418	[0.166, 0.317]	0.0974	5	0

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	0.01	0.256	[0.21, 0.335]	0.0889	5	0
tropical	seq2seq	0.1	0.3661	[0.279, 0.446]	0.104	5	0
tropical	seq2seq	1	0.3294	[0.262, 0.413]	0.1	5	0
tropical	seq2seq	10	0.2566	[0.183, 0.319]	0.0853	5	0
tropical	seq2seq	100	0.2328	[0.164, 0.297]	0.0879	5	0
tropical	seq2seq	1000	0.3208	[0.238, 0.405]	0.112	5	0
tropical	seq2seq	1e+04	0.4987	[0.46, 0.532]	0.0455	5	0
vanilla	pointer	0	0.2255	[0.139, 0.316]	0.117	5	0
vanilla	pointer	0.01	0.2297	[0.18, 0.305]	0.0855	5	0
vanilla	pointer	0.1	0.2344	[0.187, 0.305]	0.0803	5	0
vanilla	pointer	1	0.2263	[0.182, 0.294]	0.0757	5	0
vanilla	pointer	10	0.2163	[0.173, 0.277]	0.0673	5	0
vanilla	pointer	100	0.2381	[0.191, 0.304]	0.0758	5	0
vanilla	pointer	1000	0.2433	[0.187, 0.317]	0.0884	5	0
vanilla	pointer	1e+04	0.2415	[0.181, 0.315]	0.0818	5	0
vanilla	seq2seq	0	0.4899	[0.449, 0.531]	0.0525	5	0
vanilla	seq2seq	0.01	0.4631	[0.401, 0.525]	0.0806	5	0
vanilla	seq2seq	0.1	0.454	[0.385, 0.526]	0.0913	5	0
vanilla	seq2seq	1	0.4806	[0.426, 0.535]	0.068	5	0
vanilla	seq2seq	10	0.49	[0.426, 0.544]	0.075	5	0
vanilla	seq2seq	100	0.4906	[0.445, 0.538]	0.06	5	0
vanilla	seq2seq	1000	0.5072	[0.478, 0.535]	0.0409	5	0
vanilla	seq2seq	1e+04	0.454	[0.37, 0.535]	0.109	5	0

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-0.03838	[-0.204, 0.128]	0.9816
adaptive	pointer	0.1	-0.02904	[-0.195, 0.137]	0.9962
adaptive	pointer	1	-0.03958	[-0.206, 0.126]	0.9784
adaptive	pointer	10	-0.03448	[-0.201, 0.132]	0.9898
adaptive	pointer	100	-0.0304	[-0.196, 0.136]	0.9950
adaptive	pointer	1000	-0.03148	[-0.198, 0.135]	0.9939
adaptive	pointer	1e+04	-0.01738	[-0.183, 0.149]	0.9999
adaptive	seq2seq	0.01	0.00876	[-0.11, 0.127]	1.0000
adaptive	seq2seq	0.1	0.00374	[-0.115, 0.122]	1.0000
adaptive	seq2seq	1	0.00942	[-0.109, 0.128]	1.0000
adaptive	seq2seq	10	-0.01492	[-0.133, 0.103]	0.9995
adaptive	seq2seq	100	-0.0125	[-0.131, 0.106]	0.9998
adaptive	seq2seq	1000	0.01926	[-0.099, 0.138]	0.9975
adaptive	seq2seq	1e+04	0.02386	[-0.0944, 0.142]	0.9913
tropical	pointer	0.01	-0.03208	[-0.122, 0.0578]	0.8622
tropical	pointer	0.1	-0.04214	[-0.132, 0.0477]	0.6625
tropical	pointer	1	-0.03154	[-0.121, 0.0583]	0.8710
tropical	pointer	10	-0.03604	[-0.126, 0.0538]	0.7903
tropical	pointer	100	-0.03166	[-0.122, 0.0582]	0.8690
tropical	pointer	1000	-0.01872	[-0.109, 0.0712]	0.9896
tropical	pointer	1e+04	-0.03402	[-0.124, 0.0559]	0.8286
tropical	seq2seq	0.01	0.01414	[-0.147, 0.175]	1.0000
tropical	seq2seq	0.1	0.1242	[-0.0365, 0.285]	0.1855
tropical	seq2seq	1	0.0876	[-0.0732, 0.248]	0.5149
tropical	seq2seq	10	0.01478	[-0.146, 0.176]	0.9999
tropical	seq2seq	100	-0.00902	[-0.17, 0.152]	1.0000

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p	
tropical	seq2seq	1000	0.079	[-0.0818, 0.24]	0.6182	
tropical	seq2seq	1e+04	0.2568	[0.0961, 0.418]	0.0007	*
vanilla	pointer	0.01	0.00422	[-0.144, 0.153]	1.0000	
vanilla	pointer	0.1	0.0089	[-0.14, 0.157]	1.0000	
vanilla	pointer	1	8.400×10^{-4}	[-0.148, 0.149]	1.0000	
vanilla	pointer	10	-0.00914	[-0.158, 0.139]	1.0000	
vanilla	pointer	100	0.0126	[-0.136, 0.161]	1.0000	
vanilla	pointer	1000	0.0178	[-0.131, 0.166]	0.9996	
vanilla	pointer	1e+04	0.01606	[-0.133, 0.165]	0.9998	
vanilla	seq2seq	0.01	-0.02686	[-0.158, 0.104]	0.9905	
vanilla	seq2seq	0.1	-0.03598	[-0.167, 0.0951]	0.9560	
vanilla	seq2seq	1	-0.00934	[-0.14, 0.122]	1.0000	
vanilla	seq2seq	10	4.000×10^{-5}	[-0.131, 0.131]	1.0000	
vanilla	seq2seq	100	6.200×10^{-4}	[-0.13, 0.132]	1.0000	
vanilla	seq2seq	1000	0.01722	[-0.114, 0.148]	0.9994	
vanilla	seq2seq	1e+04	-0.03596	[-0.167, 0.0951]	0.9561	

Table 23: `ood_noise_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.002404	[-0.00321, 0.00597]	0.7371	0.0876	0.5183
adaptive	seq2seq	0.001944	[-0.00705, 0.0151]	0.7225	-0.0571	0.6755
tropical	pointer	0.001461	[-0.00312, 0.00634]	0.7138	-0.0019	0.9910
tropical	seq2seq	0.01932	[0.00262, 0.0346]	0.0535	0.272	0.0400
vanilla	pointer	0.002324	$[-6.16 \times 10^{-4}$, 0.00477]	0.7112	0.0819	0.5495
vanilla	seq2seq	0.003181	[-0.00159, 0.00864]	0.6142	0.0171	0.9102

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	-0.0149	[-0.209, 0.179]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-0.0512	[-0.226, 0.124]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.06594	[-0.233, 0.101]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	-0.04728	[-0.185, 0.0901]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	-0.0418	[-0.16, 0.0765]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	-0.05916	[-0.207, 0.089]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.05142	[-0.185, 0.0817]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-0.06498	[-0.213, 0.0831]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.2481	[-0.373, -0.124]	0.0785

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.2071	[-0.296, -0.118]	0.0475	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.0879	[-0.187, 0.0115]	0.7004	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.1512	[-0.246, -0.0562]	0.1496	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.2334	[-0.386, -0.0811]	0.1572	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.2578	[-0.4, -0.115]	0.1029	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.1863	[-0.371, -0.00171]	0.5357	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.04468	[-0.0985, 0.188]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.0425	[-0.089, 0.174]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-1.000×10^{-4}	[-0.0735, 0.0733]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.00456	[-0.0644, 0.0735]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.00208	[-0.0735, 0.0776]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.01716	[-0.0709, 0.105]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-5.000×10^{-4}	[-0.0715, 0.0705]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.00678	[-0.0838, 0.0702]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.00906	[-0.0788, 0.0969]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.00574	[-0.0512, 0.0627]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.04136	[-0.0642, 0.147]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.04546	[-0.0523, 0.143]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.0245	[-0.0469, 0.0959]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.00922	[-0.0675, 0.0491]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.00738	[-0.0889, 0.0741]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.00778	[-0.0144, 0.03]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.06556	[-0.0339, 0.165]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.2277	[-0.373, -0.0822]	0.0977	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.2748	[-0.375, -0.175]	0.0274	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.2605	[-0.358, -0.163]	0.0274	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.2767	[-0.378, -0.176]	0.0274	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.2473	[-0.307, -0.187]	0.0080	*

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.2456	[-0.334, -0.157]	0.0274	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.2784	[-0.353, -0.204]	0.0114	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.2689	[-0.39, -0.148]	0.0420	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.03124	[-0.163, 0.101]	0.5465	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.07746	[-0.186, 0.0315]	0.4784	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.1976	[-0.369, -0.0259]	0.2140	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.1504	[-0.318, 0.0174]	0.3381	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.08206	[-0.215, 0.051]	0.4859	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.05388	[-0.149, 0.0408]	0.4859	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.129	[-0.238, -0.0197]	0.2140	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	-0.3221	[-0.414, -0.231]	0.0136	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.2645	[-0.412, -0.117]	0.0686	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.2334	[-0.329, -0.137]	0.0326	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.2196	[-0.322, -0.117]	0.0438	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.2543	[-0.345, -0.164]	0.0274	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.2736	[-0.36, -0.187]	0.0194	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.2525	[-0.334, -0.171]	0.0205	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.2639	[-0.363, -0.165]	0.0274	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.2124	[-0.32, -0.105]	0.0530	

A.6 `id_f1`

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.5135	[0.495, 0.525]	0.0211	5	0
adaptive	pointer	0.01	0.5207	[0.501, 0.533]	0.0221	5	0
adaptive	pointer	0.1	0.5195	[0.5, 0.531]	0.0214	5	0
adaptive	pointer	1	0.5202	[0.502, 0.531]	0.0208	5	0
adaptive	pointer	10	0.5208	[0.502, 0.532]	0.0213	5	0
adaptive	pointer	100	0.5202	[0.499, 0.532]	0.0232	5	0
adaptive	pointer	1000	0.5218	[0.503, 0.534]	0.0211	5	0
adaptive	pointer	1e+04	0.5199	[0.501, 0.53]	0.0205	5	0
adaptive	seq2seq	0	0.3822	[0.347, 0.408]	0.0401	5	0
adaptive	seq2seq	0.01	0.3846	[0.351, 0.407]	0.0377	5	0

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	seq2seq	0.1	0.3815	[0.355, 0.401]	0.0309	5	0
adaptive	seq2seq	1	0.3814	[0.353, 0.404]	0.0337	5	0
adaptive	seq2seq	10	0.3815	[0.349, 0.404]	0.0369	5	0
adaptive	seq2seq	100	0.3724	[0.334, 0.4]	0.0442	5	0
adaptive	seq2seq	1000	0.2629	[0.223, 0.302]	0.0522	5	0
adaptive	seq2seq	1e+04	0.1024	[0.0467, 0.151]	0.067	5	0
tropical	pointer	0	0.5563	[0.534, 0.578]	0.0277	5	0
tropical	pointer	0.01	0.5694	[0.555, 0.586]	0.0199	5	0
tropical	pointer	0.1	0.5607	[0.546, 0.577]	0.0211	5	0
tropical	pointer	1	0.5699	[0.552, 0.585]	0.022	5	0
tropical	pointer	10	0.5644	[0.554, 0.577]	0.0146	5	0
tropical	pointer	100	0.5672	[0.554, 0.58]	0.0179	5	0
tropical	pointer	1000	0.5661	[0.551, 0.584]	0.0216	5	0
tropical	pointer	1e+04	0.5604	[0.544, 0.577]	0.0219	5	0
tropical	seq2seq	0	0.5031	[0.479, 0.525]	0.0296	5	0
tropical	seq2seq	0.01	0.4995	[0.475, 0.518]	0.0289	5	0
tropical	seq2seq	0.1	0.5021	[0.473, 0.525]	0.0341	5	0
tropical	seq2seq	1	0.4947	[0.464, 0.524]	0.0367	5	0
tropical	seq2seq	10	0.5	[0.477, 0.517]	0.0255	5	0
tropical	seq2seq	100	0.4926	[0.479, 0.502]	0.0159	5	0
tropical	seq2seq	1000	0.4317	[0.402, 0.467]	0.0409	5	0
tropical	seq2seq	1e+04	0.0864	[0.0211, 0.155]	0.0855	5	0
vanilla	pointer	0	0.5072	[0.488, 0.526]	0.0252	5	0
vanilla	pointer	0.01	0.5183	[0.502, 0.535]	0.0218	5	0
vanilla	pointer	0.1	0.518	[0.501, 0.535]	0.022	5	0
vanilla	pointer	1	0.5182	[0.501, 0.535]	0.0226	5	0
vanilla	pointer	10	0.5173	[0.501, 0.534]	0.0224	5	0
vanilla	pointer	100	0.5175	[0.501, 0.534]	0.0221	5	0
vanilla	pointer	1000	0.5171	[0.499, 0.536]	0.0248	5	0
vanilla	pointer	1e+04	0.5185	[0.502, 0.535]	0.0222	5	0
vanilla	seq2seq	0	0.3839	[0.344, 0.415]	0.0467	5	0
vanilla	seq2seq	0.01	0.3827	[0.349, 0.413]	0.0409	5	0
vanilla	seq2seq	0.1	0.3813	[0.351, 0.412]	0.0423	5	0
vanilla	seq2seq	1	0.3829	[0.349, 0.413]	0.0404	5	0
vanilla	seq2seq	10	0.3887	[0.349, 0.417]	0.0461	5	0
vanilla	seq2seq	100	0.3828	[0.336, 0.413]	0.0505	5	0
vanilla	seq2seq	1000	0.2764	[0.25, 0.304]	0.0336	5	0
vanilla	seq2seq	1e+04	0.1218	[0.0684, 0.159]	0.0591	5	0

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0.007179	[-0.0303, 0.0447]	0.9936
adaptive	pointer	0.1	0.00593	[-0.0316, 0.0434]	0.9979
adaptive	pointer	1	0.006643	[-0.0308, 0.0441]	0.9959
adaptive	pointer	10	0.007283	[-0.0302, 0.0448]	0.9930
adaptive	pointer	100	0.006625	[-0.0309, 0.0441]	0.9960
adaptive	pointer	1000	0.008216	[-0.0293, 0.0457]	0.9862
adaptive	pointer	1e+04	0.006318	[-0.0312, 0.0438]	0.9970
adaptive	seq2seq	0.01	0.002401	[-0.0749, 0.0797]	1.0000
adaptive	seq2seq	0.1	-6.942×10^{-4}	[-0.0779, 0.0766]	1.0000
adaptive	seq2seq	1	-8.149×10^{-4}	[-0.0781, 0.0764]	1.0000
adaptive	seq2seq	10	-6.654×10^{-4}	[-0.0779, 0.0766]	1.0000

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p	
adaptive	seq2seq	100	-0.00976	[-0.087, 0.0675]	0.9995	
adaptive	seq2seq	1000	-0.1193	[-0.197, -0.0421]	0.0010	*
adaptive	seq2seq	1e+04	-0.2798	[-0.357, -0.203]	$< 10^{-4}$	*
tropical	pointer	0.01	0.01314	[-0.0238, 0.0501]	0.8635	
tropical	pointer	0.1	0.004439	[-0.0325, 0.0414]	0.9996	
tropical	pointer	1	0.01364	[-0.0233, 0.0506]	0.8430	
tropical	pointer	10	0.008164	[-0.0288, 0.0451]	0.9855	
tropical	pointer	100	0.01093	[-0.026, 0.0478]	0.9372	
tropical	pointer	1000	0.009839	[-0.0271, 0.0468]	0.9618	
tropical	pointer	1e+04	0.004186	[-0.0327, 0.0411]	0.9998	
tropical	seq2seq	0.01	-0.003576	[-0.0769, 0.0698]	1.0000	
tropical	seq2seq	0.1	-9.973×10^{-4}	[-0.0743, 0.0724]	1.0000	
tropical	seq2seq	1	-0.008355	[-0.0817, 0.065]	0.9997	
tropical	seq2seq	10	-0.003091	[-0.0764, 0.0703]	1.0000	
tropical	seq2seq	100	-0.01049	[-0.0838, 0.0629]	0.9989	
tropical	seq2seq	1000	-0.07138	[-0.145, 0.00197]	0.0591	
tropical	seq2seq	1e+04	-0.4167	[-0.49, -0.343]	$< 10^{-4}$	*
vanilla	pointer	0.01	0.01113	[-0.0289, 0.0512]	0.9533	
vanilla	pointer	0.1	0.01079	[-0.0293, 0.0508]	0.9600	
vanilla	pointer	1	0.01094	[-0.0291, 0.051]	0.9570	
vanilla	pointer	10	0.01008	[-0.03, 0.0501]	0.9714	
vanilla	pointer	100	0.01024	[-0.0298, 0.0503]	0.9690	
vanilla	pointer	1000	0.009875	[-0.0302, 0.0499]	0.9742	
vanilla	pointer	1e+04	0.01131	[-0.0288, 0.0514]	0.9498	
vanilla	seq2seq	0.01	-0.001203	[-0.0807, 0.0783]	1.0000	
vanilla	seq2seq	0.1	-0.002582	[-0.0821, 0.0769]	1.0000	
vanilla	seq2seq	1	-0.001009	[-0.0805, 0.0785]	1.0000	
vanilla	seq2seq	10	0.004793	[-0.0747, 0.0843]	1.0000	
vanilla	seq2seq	100	-0.001145	[-0.0806, 0.0783]	1.0000	
vanilla	seq2seq	1000	-0.1075	[-0.187, -0.028]	0.0044	*
vanilla	seq2seq	1e+04	-0.2621	[-0.342, -0.183]	$< 10^{-4}$	*

Table 27: `id_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	7.045×10^{-5}	$[-1.74 \times 10^{-4}, 3.59 \times 10^{-4}]$	0.9667	0.00952	0.9544
adaptive	seq2seq	-0.03903	[-0.0429, -0.0346]	$< 10^{-4}$	-0.558	0.0002
tropical	pointer	-6.706×10^{-4}	$[-0.00171, 3.73 \times 10^{-4}]$	0.6748	-0.0362	0.7938
tropical	seq2seq	-0.04936	[-0.0575, -0.0402]	$< 10^{-4}$	-0.558	0.0002
vanilla	pointer	-7.105×10^{-5}	$[-3.49 \times 10^{-4}, 2.07 \times 10^{-4}]$	0.9680	-0.00952	0.9556
vanilla	seq2seq	-0.03545	[-0.0405, -0.0299]	$< 10^{-4}$	-0.493	0.0002

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.04904	$[-5.79 \times 10^{-4}, 0.0987]$	0.1655
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.05106	[0.0128, 0.0893]	0.1393

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.0427	[0.00181, 0.0836]	0.1655	
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.05174	[0.0146, 0.0889]	0.1393	
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.04713	[0.0146, 0.0796]	0.1393	
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.04973	[0.0156, 0.0838]	0.1393	
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.04901	[0.00311, 0.0949]	0.1655	
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.04192	[0.0105, 0.0734]	0.1393	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.1192	[0.0652, 0.173]	0.0358	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.1168	[0.0741, 0.159]	0.0225	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.1207	[0.0753, 0.166]	0.0235	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.1118	[0.0649, 0.159]	0.0324	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.1113	[0.0747, 0.148]	0.0162	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.1098	[0.063, 0.157]	0.0324	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.1553	[0.119, 0.191]	0.0045	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.03542	[-0.135, 0.0643]	0.3797	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.006318	[-0.0132, 0.0258]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.002366	[-0.0225, 0.0272]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.001463	[-0.0222, 0.0252]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.002021	[-0.0211, 0.0251]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.00352	[-0.0208, 0.0278]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.0027	[-0.0211, 0.0265]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.004659	[-0.0178, 0.0271]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.00133	[-0.0227, 0.0254]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.001708	[-0.0304, 0.027]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.001896	[-0.0212, 0.025]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	1.799×10^{-4}	[-0.0186, 0.019]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.001514	[-0.0176, 0.0146]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.007167	[-0.0263, 0.012]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.01032	[-0.0431, 0.0225]	1.0000	

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.01348	[-0.0437, 0.0167]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.0194	[-0.0519, 0.0131]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.1313	[0.074, 0.189]	0.0294	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.1361	[0.0792, 0.193]	0.0294	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.138	[0.0907, 0.185]	0.0240	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.1388	[0.088, 0.19]	0.0281	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.1393	[0.0839, 0.195]	0.0292	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.1477	[0.0849, 0.211]	0.0294	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.2589	[0.176, 0.341]	0.0190	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.4174	[0.32, 0.515]	0.0064	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.0532	[0.0192, 0.0872]	0.0438	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.06991	[0.0297, 0.11]	0.0438	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.05863	[0.00974, 0.108]	0.0513	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.07519	[0.015, 0.135]	0.0513	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.06445	[0.0333, 0.0956]	0.0365	*
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.07461	[0.0476, 0.102]	0.0281	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.1344	[0.0819, 0.187]	0.0292	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.474	[0.359, 0.589]	0.0069	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.1233	[0.0553, 0.191]	0.0438	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.1356	[0.0859, 0.185]	0.0281	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.1367	[0.0823, 0.191]	0.0292	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.1353	[0.0833, 0.187]	0.0292	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.1286	[0.0644, 0.193]	0.0365	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.1347	[0.0561, 0.213]	0.0438	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.2407	[0.189, 0.293]	0.0051	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.3967	[0.308, 0.485]	0.0055	*

A.7 id_exact

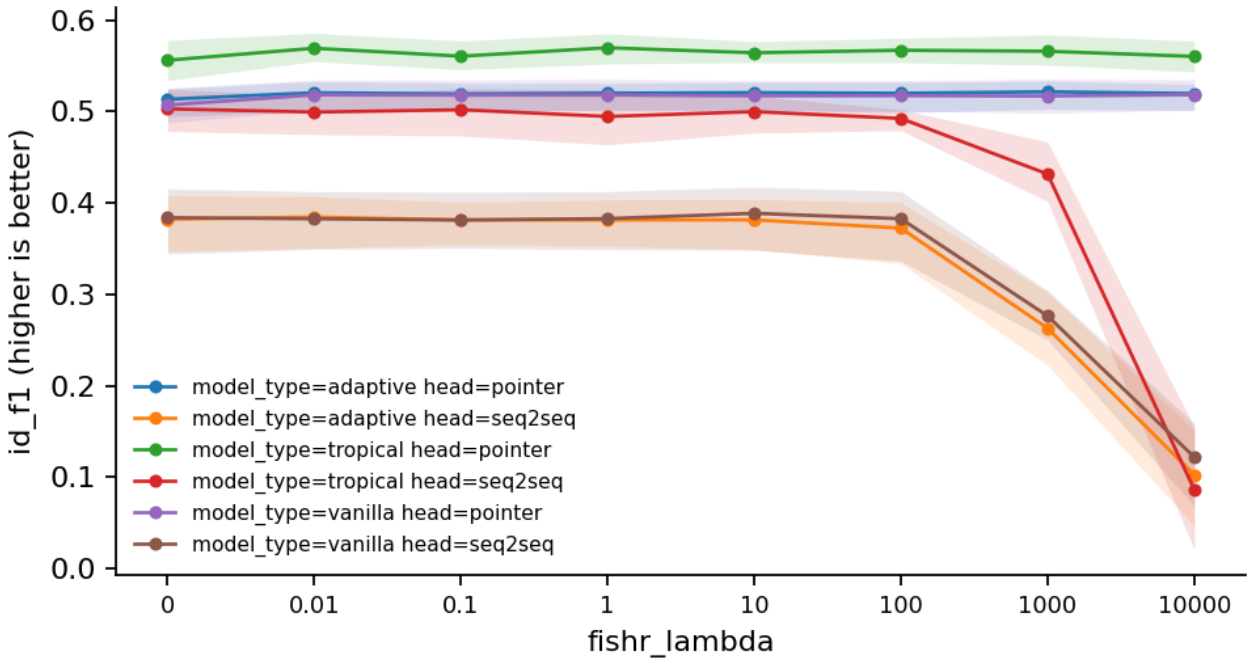


Figure 7: `id_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

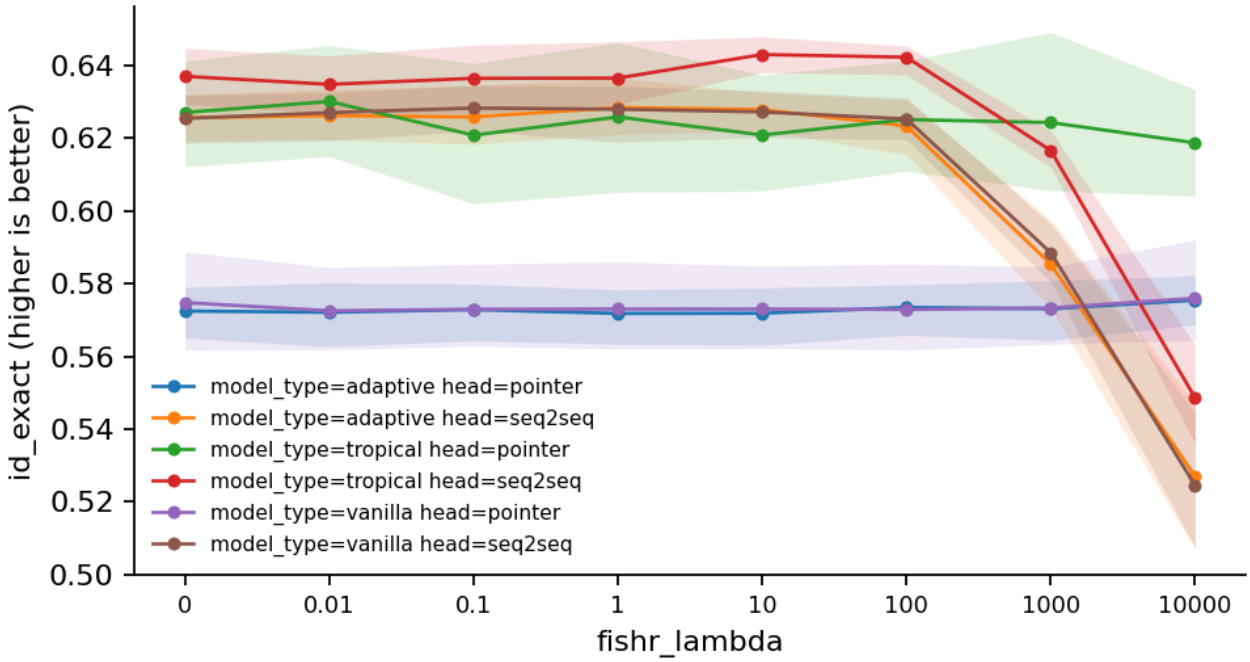


Figure 8: `id_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 29: `id_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.5724	[0.565, 0.579]	0.00944	5	0
adaptive	pointer	0.01	0.5721	[0.563, 0.58]	0.0111	5	0
adaptive	pointer	0.1	0.5728	[0.564, 0.58]	0.00986	5	0
adaptive	pointer	1	0.5718	[0.563, 0.578]	0.00975	5	0
adaptive	pointer	10	0.5718	[0.563, 0.579]	0.0102	5	0
adaptive	pointer	100	0.5734	[0.566, 0.58]	0.00894	5	0
adaptive	pointer	1000	0.5731	[0.564, 0.581]	0.00999	5	0
adaptive	pointer	1e+04	0.5754	[0.569, 0.582]	0.00901	5	0
adaptive	seq2seq	0	0.6256	[0.619, 0.632]	0.00769	5	0
adaptive	seq2seq	0.01	0.6262	[0.62, 0.633]	0.00881	5	0
adaptive	seq2seq	0.1	0.6258	[0.618, 0.634]	0.00997	5	0
adaptive	seq2seq	1	0.6284	[0.621, 0.637]	0.00987	5	0
adaptive	seq2seq	10	0.6278	[0.622, 0.633]	0.00677	5	0
adaptive	seq2seq	100	0.6233	[0.615, 0.63]	0.00972	5	0
adaptive	seq2seq	1000	0.5853	[0.573, 0.598]	0.0166	5	0
adaptive	seq2seq	1e+04	0.527	[0.507, 0.547]	0.0275	5	0
tropical	pointer	0	0.627	[0.612, 0.641]	0.0186	5	0
tropical	pointer	0.01	0.63	[0.615, 0.645]	0.0197	5	0
tropical	pointer	0.1	0.6208	[0.602, 0.64]	0.0262	5	0
tropical	pointer	1	0.6258	[0.605, 0.646]	0.0277	5	0
tropical	pointer	10	0.6208	[0.605, 0.637]	0.0205	5	0
tropical	pointer	100	0.6251	[0.611, 0.641]	0.0191	5	0
tropical	pointer	1000	0.6243	[0.606, 0.649]	0.0266	5	0
tropical	pointer	1e+04	0.6187	[0.604, 0.633]	0.0184	5	0
tropical	seq2seq	0	0.637	[0.629, 0.645]	0.01	5	0
tropical	seq2seq	0.01	0.6348	[0.627, 0.643]	0.0101	5	0
tropical	seq2seq	0.1	0.6364	[0.627, 0.646]	0.0123	5	0
tropical	seq2seq	1	0.6365	[0.629, 0.646]	0.0112	5	0
tropical	seq2seq	10	0.643	[0.638, 0.648]	0.0064	5	0
tropical	seq2seq	100	0.6422	[0.637, 0.645]	0.00548	5	0
tropical	seq2seq	1000	0.6166	[0.612, 0.622]	0.00649	5	0
tropical	seq2seq	1e+04	0.5486	[0.536, 0.563]	0.0183	5	0
vanilla	pointer	0	0.5748	[0.562, 0.589]	0.0169	5	0
vanilla	pointer	0.01	0.5725	[0.562, 0.584]	0.0147	5	0
vanilla	pointer	0.1	0.5729	[0.563, 0.585]	0.0154	5	0
vanilla	pointer	1	0.573	[0.562, 0.586]	0.0154	5	0
vanilla	pointer	10	0.573	[0.562, 0.585]	0.015	5	0
vanilla	pointer	100	0.5729	[0.562, 0.585]	0.0151	5	0
vanilla	pointer	1000	0.5733	[0.563, 0.585]	0.0143	5	0
vanilla	pointer	1e+04	0.576	[0.564, 0.592]	0.0186	5	0
vanilla	seq2seq	0	0.6254	[0.619, 0.632]	0.0079	5	0
vanilla	seq2seq	0.01	0.627	[0.619, 0.633]	0.00859	5	0
vanilla	seq2seq	0.1	0.6283	[0.622, 0.635]	0.00767	5	0
vanilla	seq2seq	1	0.6279	[0.619, 0.634]	0.0097	5	0
vanilla	seq2seq	10	0.6272	[0.62, 0.633]	0.00843	5	0
vanilla	seq2seq	100	0.6252	[0.62, 0.631]	0.00726	5	0
vanilla	seq2seq	1000	0.5885	[0.581, 0.596]	0.0106	5	0
vanilla	seq2seq	1e+04	0.5246	[0.507, 0.545]	0.0241	5	0

Table 30: `id_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	-3.400×10^{-4}	[-0.0175, 0.0168]	1.0000	
adaptive	pointer	0.1	4.000×10^{-4}	[-0.0167, 0.0175]	1.0000	
adaptive	pointer	1	-6.600×10^{-4}	[-0.0178, 0.0165]	1.0000	
adaptive	pointer	10	-6.000×10^{-4}	[-0.0177, 0.0165]	1.0000	
adaptive	pointer	100	0.001	[-0.0161, 0.0181]	1.0000	
adaptive	pointer	1000	6.200×10^{-4}	[-0.0165, 0.0177]	1.0000	
adaptive	pointer	1e+04	0.003	[-0.0141, 0.0201]	0.9962	
adaptive	seq2seq	0.01	5.800×10^{-4}	[-0.0234, 0.0245]	1.0000	
adaptive	seq2seq	0.1	2.200×10^{-4}	[-0.0237, 0.0242]	1.0000	
adaptive	seq2seq	1	0.00282	[-0.0211, 0.0268]	0.9997	
adaptive	seq2seq	10	0.00224	[-0.0217, 0.0262]	0.9999	
adaptive	seq2seq	100	-0.00226	[-0.0262, 0.0217]	0.9999	
adaptive	seq2seq	1000	-0.04024	[-0.0642, -0.0163]	0.0003	*
adaptive	seq2seq	1e+04	-0.09856	[-0.123, -0.0746]	$< 10^{-4}$	*
tropical	pointer	0.01	0.003	[-0.0362, 0.0422]	1.0000	
tropical	pointer	0.1	-0.00626	[-0.0454, 0.0329]	0.9978	
tropical	pointer	1	-0.00124	[-0.0404, 0.0379]	1.0000	
tropical	pointer	10	-0.00622	[-0.0454, 0.0329]	0.9979	
tropical	pointer	100	-0.00196	[-0.0411, 0.0372]	1.0000	
tropical	pointer	1000	-0.00278	[-0.0419, 0.0364]	1.0000	
tropical	pointer	1e+04	-0.00836	[-0.0475, 0.0308]	0.9881	
tropical	seq2seq	0.01	-0.0022	[-0.0211, 0.0167]	0.9997	
tropical	seq2seq	0.1	-5.400×10^{-4}	[-0.0194, 0.0183]	1.0000	
tropical	seq2seq	1	-4.800×10^{-4}	[-0.0193, 0.0184]	1.0000	
tropical	seq2seq	10	0.006	[-0.0129, 0.0249]	0.9136	
tropical	seq2seq	100	0.00526	[-0.0136, 0.0241]	0.9524	
tropical	seq2seq	1000	-0.02038	[-0.0392, -0.00153]	0.0296	*
tropical	seq2seq	1e+04	-0.08836	[-0.107, -0.0695]	$< 10^{-4}$	*
vanilla	pointer	0.01	-0.00224	[-0.0297, 0.0252]	1.0000	
vanilla	pointer	0.1	-0.00184	[-0.0293, 0.0256]	1.0000	
vanilla	pointer	1	-0.00174	[-0.0292, 0.0257]	1.0000	
vanilla	pointer	10	-0.00176	[-0.0292, 0.0257]	1.0000	
vanilla	pointer	100	-0.0019	[-0.0294, 0.0256]	1.0000	
vanilla	pointer	1000	-0.00148	[-0.029, 0.026]	1.0000	
vanilla	pointer	1e+04	0.00122	[-0.0263, 0.0287]	1.0000	
vanilla	seq2seq	0.01	0.00158	[-0.019, 0.0221]	1.0000	
vanilla	seq2seq	0.1	0.00286	[-0.0177, 0.0234]	0.9991	
vanilla	seq2seq	1	0.00252	[-0.018, 0.0231]	0.9996	
vanilla	seq2seq	10	0.00178	[-0.0188, 0.0223]	1.0000	
vanilla	seq2seq	100	-1.600×10^{-4}	[-0.0207, 0.0204]	1.0000	
vanilla	seq2seq	1000	-0.03692	[-0.0575, -0.0164]	0.0001	*
vanilla	seq2seq	1e+04	-0.1008	[-0.121, -0.0803]	$< 10^{-4}$	*

Table 31: `id_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	4.329×10^{-4}	$[1.03 \times 10^{-4}, 7.71 \times 10^{-4}]$	0.5779	0.0838	0.5395
adaptive	seq2seq	-0.01369	[-0.0169, -0.0105]	$< 10^{-4}$	-0.531	0.0002
tropical	pointer	-9.943×10^{-4}	$[-0.00174, -2.50 \times 10^{-4}]$	0.5848	-0.0248	0.8484
tropical	seq2seq	-0.01044	[-0.0125, -0.00838]	$< 10^{-4}$	-0.392	0.0030
vanilla	pointer	3.907×10^{-4}	$[-3.71 \times 10^{-5}, 8.76 \times 10^{-4}]$	0.7493	0.0419	0.7658

Table 31: `id_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
vanilla	seq2seq	-0.01391	[-0.0166, -0.0112]	$< 10^{-4}$	-0.573	0.0002

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.05228	[0.0269, 0.0776]	0.0692
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.05752	[0.0307, 0.0843]	0.0639
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.04786	[0.0102, 0.0855]	0.1450
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.05278	[0.017, 0.0886]	0.1363
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.04782	[0.0202, 0.0754]	0.1030
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.05222	[0.0239, 0.0805]	0.0961
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.05098	[0.0147, 0.0873]	0.1363
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.0427	[0.0195, 0.0659]	0.0961
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.01156	[-0.00795, 0.0311]	0.8767
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.00778	[-0.0125, 0.0281]	0.8767
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.00816	[-0.0142, 0.0306]	0.8767
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.00856	[-0.00722, 0.0243]	0.8767
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.01578	[0.00495, 0.0266]	0.1363
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.01698	[0.00577, 0.0282]	0.1363
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.0281	[0.0111, 0.0451]	0.1106
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.02404	[-0.0277, 0.0758]	0.8767
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.00232	[-0.0195, 0.0149]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-4.200×10^{-4}	[-0.0157, 0.0149]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-8.000×10^{-5}	[-0.0155, 0.0153]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.00124	[-0.0176, 0.0152]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.00116	[-0.0167, 0.0144]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	5.800×10^{-4}	[-0.0141, 0.0153]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-2.200×10^{-4}	[-0.0168, 0.0163]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-5.400×10^{-4}	[-0.0175, 0.0165]	1.0000

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	1.800×10^{-4}	[-0.00234, 0.0027]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-8.200×10^{-4}	[-0.00616, 0.00452]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.00246	[-0.00593, 0.00101]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	4.800×10^{-4}	[-0.00856, 0.00952]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	6.400×10^{-4}	[-0.00386, 0.00514]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.00192	[-0.00651, 0.00267]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.00314	[-0.0233, 0.0171]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.00246	[-0.00224, 0.00716]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.05314	[-0.0599, -0.0464]	0.0006	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.05406	[-0.0638, -0.0443]	0.0022	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.05296	[-0.0634, -0.0425]	0.0029	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.05662	[-0.0687, -0.0445]	0.0038	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.05598	[-0.0637, -0.0483]	0.0008	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.04988	[-0.0566, -0.0432]	0.0007	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.01228	[-0.0441, 0.0195]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.04842	[0.0137, 0.0831]	0.1973	
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.00992	[-0.035, 0.0151]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.00472	[-0.0326, 0.0232]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.01564	[-0.0512, 0.0199]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.01068	[-0.0569, 0.0356]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.02214	[-0.0489, 0.00461]	0.6650	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.01714	[-0.0388, 0.00453]	0.6650	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.00768	[-0.0303, 0.0457]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.07008	[0.0353, 0.105]	0.0604	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.05064	[-0.0668, -0.0345]	0.0133	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.05446	[-0.0691, -0.0398]	0.0079	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.05534	[-0.0675, -0.0431]	0.0041	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.0549	[-0.071, -0.0388]	0.0103	*

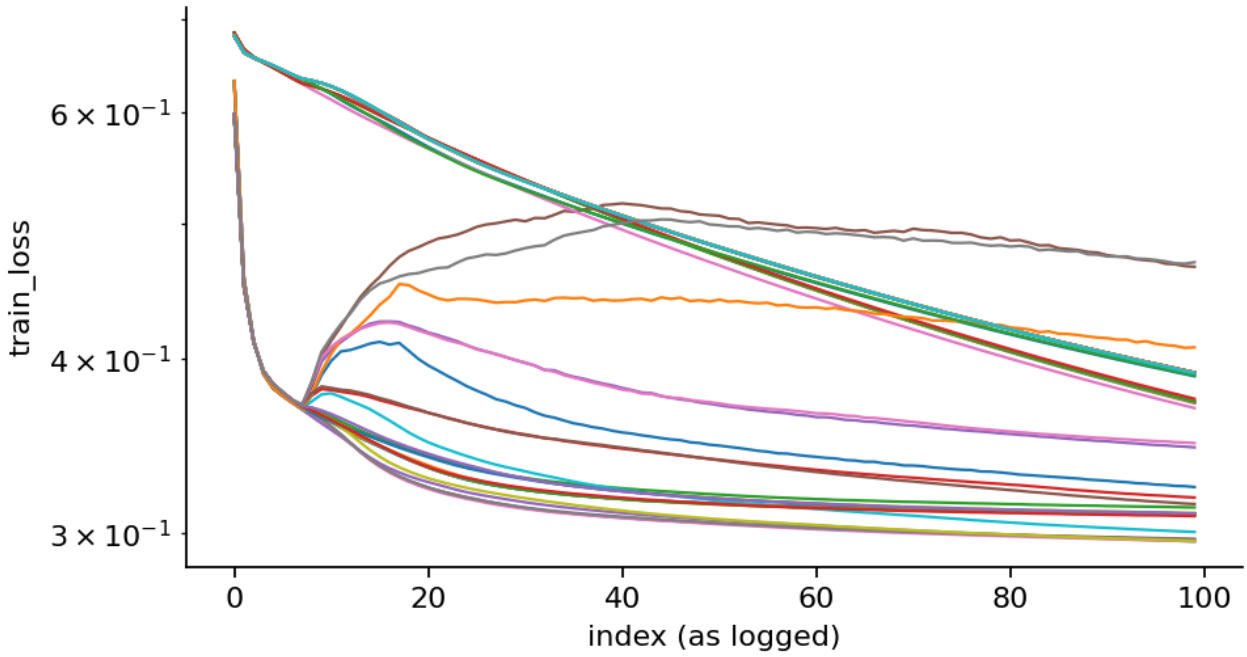


Figure 9: `train_loss`, the mean over seeds per cell.

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.05418	[-0.0728, -0.0355]	0.0167	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.05238	[-0.0646, -0.0401]	0.0049	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.0152	[-0.0308, 3.84×10^{-4}]	0.4829	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.05142	[0.0124, 0.0905]	0.2163	

B Curves

C Configuration

The base configuration every run shares (frozen hash `037f987733ac`); factor levels are in Table 1.

```
task:
  name: SubsetSumDecisionDataset
  value_range:
  - -5
  - 5
  target_range:
  - 1
  - 10
  weight_range: null
  adversarial_range:
  - 10
  - 30
  noise_prob: 0.0
```

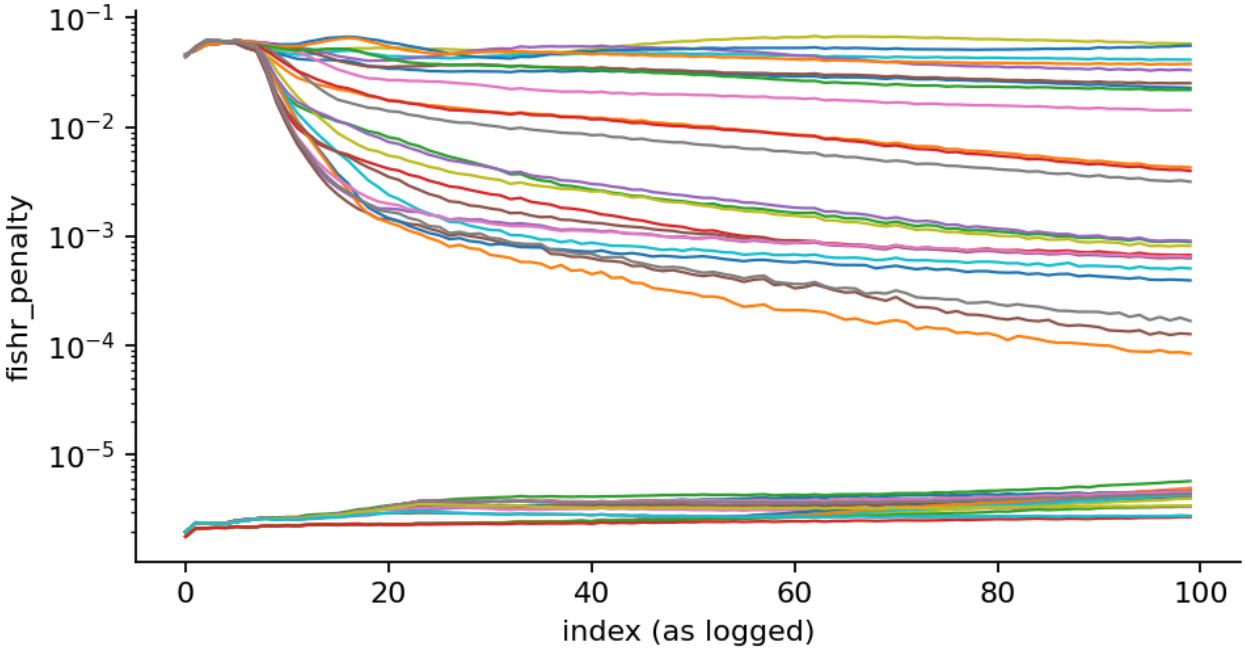


Figure 10: `fishr_penalty`, the mean over seeds per cell.

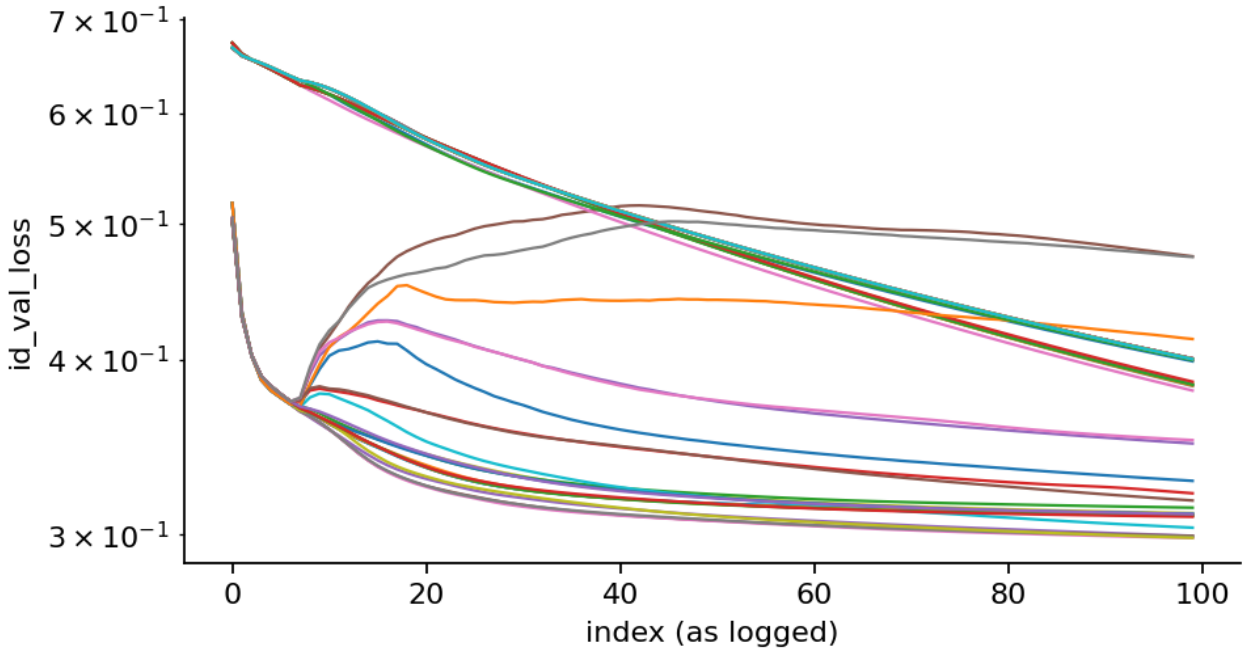


Figure 11: `id_val_loss`, the mean over seeds per cell.

```
data:
  train_lengths:
  - 4
  - 5
  - 6
  - 7
  - 8
  ood_lengths:
  - 12
  - 16
  - 64
  ood_value_range:
  - -20
  - 20
  ood_noise_range:
  - 10
  - 30
  ood_noise_prob: 0.5
  n_per_length: 20000
  n_eval_per_length: 2000
  curve_eval_limit: 256
model:
  d_model: 64
  n_heads: 2
  num_layers: 1
  dropout: 0.0
  tropical_impl: legacy
optim:
  lr: 0.0001
  epochs: 100
  batch_size: 100
  eval_batch_size: 256
fishr:
  ema: 0.95
  anneal_iters: 1500
  reset_optimizer: true
```

Store: `campaigns/tropical/SubsetSumDecision/results`.